



HA NOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

Computer Vision

Chapter 7 (part 3):
Deep Learning for CV

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Deep learning for CV



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Contents

1. Object detection: sliding-windows
2. Two-stage object detection
3. One-stage object detection: Anchor-based
4. One-stage object detection: Anchor-free
5. Semantic segmentation
6. Instance segmentation



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Computer Vision Tasks

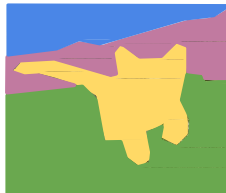
Classification



CAT

No spatial extent

Semantic Segmentation



**GRASS, CAT,
TREE, SKY**

No objects, just pixels

Object Detection



DOG, DOG, CAT

Multiple Object

Instance Segmentation



DOG, DOG, CAT



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Object detection: sliding windows



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Object Detection

Classification



CAT

No spatial extent

Semantic Segmentation



GRASS, CAT, TREE, SKY

No objects, just pixels

Object Detection



DOG, DOG, CAT

Multiple Object

Instance Segmentation



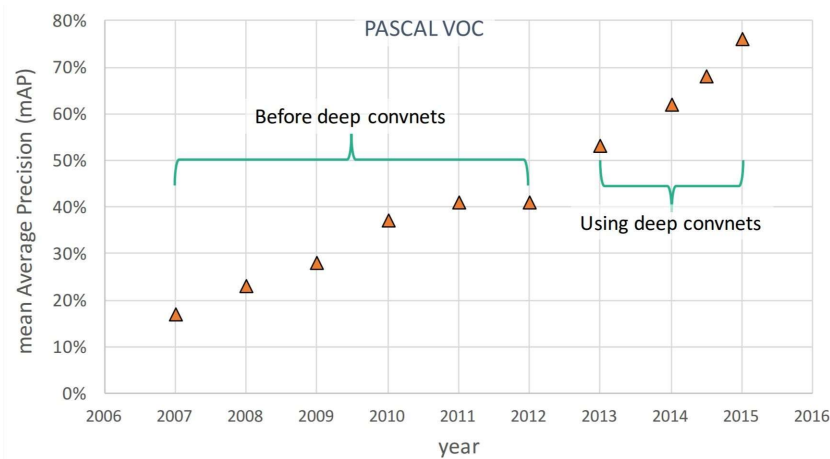
DOG, DOG, CAT



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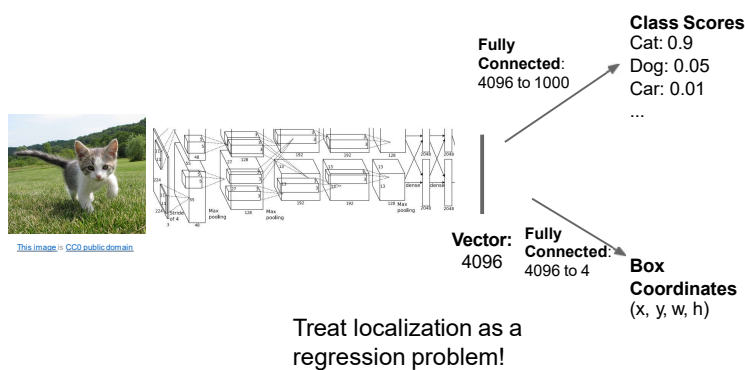
Object Detection: Impact of Deep Learning



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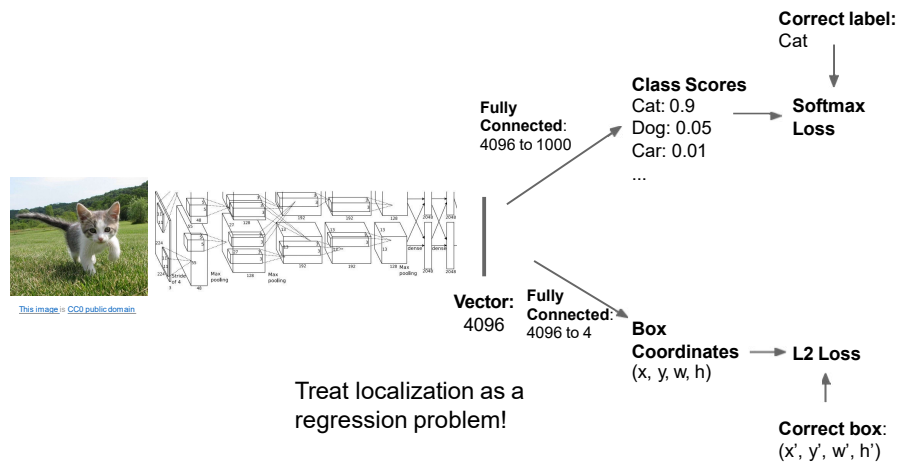
Object Detection: Single Object (Classification + Localization)



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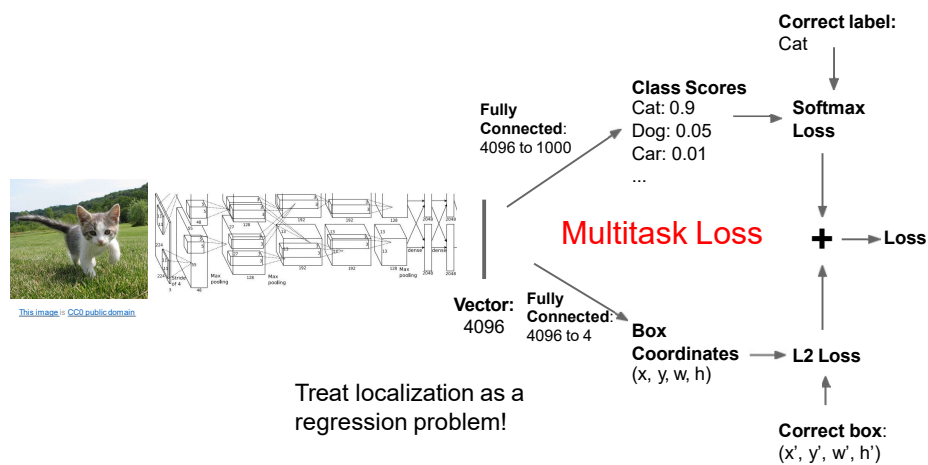
Object Detection: Single Object (Classification + Localization)



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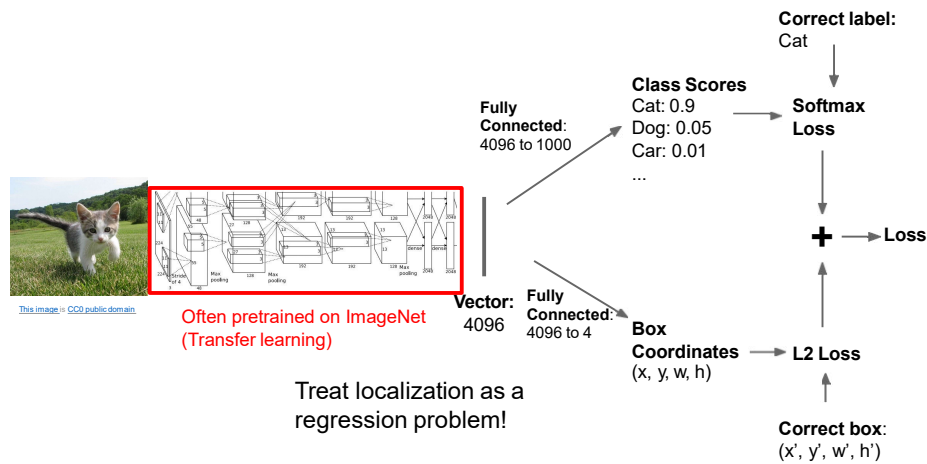
Object Detection: Single Object (Classification + Localization)



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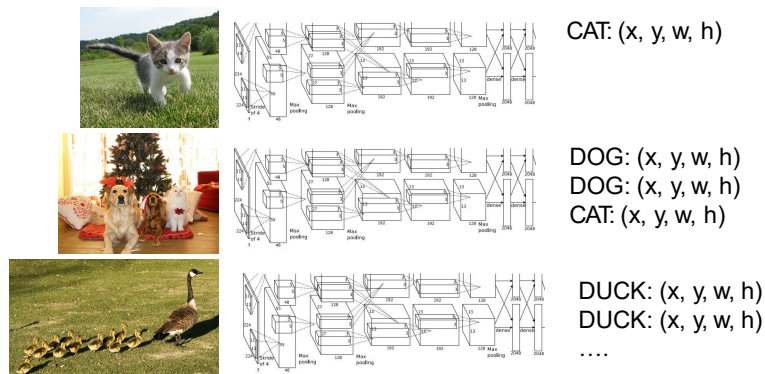
Object Detection: Single Object (Classification + Localization)



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Object Detection: Multiple Objects

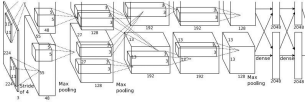
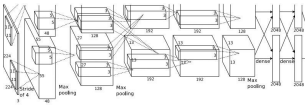
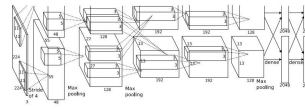


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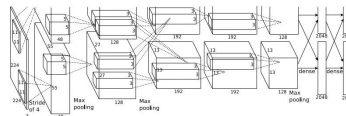
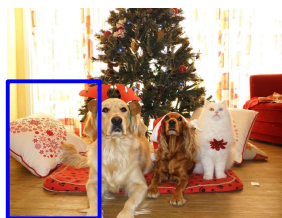
Object Detection: Multiple Objects

Each image needs a different number of outputs!

		CAT: (x, y, w, h)	4 numbers
		DOG: (x, y, w, h) DOG: (x, y, w, h) CAT: (x, y, w, h)	16 numbers
		DUCK: (x, y, w, h) DUCK: (x, y, w, h)	Many numbers!

Object Detection: Multiple Objects

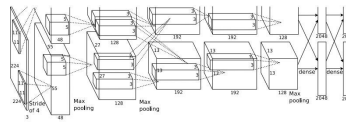
Apply a CNN to many different crops of the image, CNN classifies each crop as object or background



Dog? NO
Cat? NO
Background? YES

Object Detection: Multiple Objects

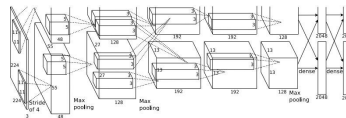
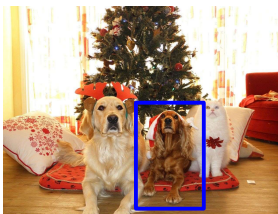
Apply a CNN to many different crops of the image, CNN classifies each crop as object or background



Dog? YES
Cat? NO
Background? NO

Object Detection: Multiple Objects

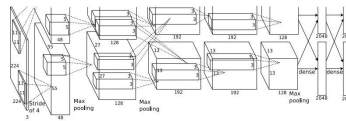
Apply a CNN to many different crops of the image, CNN classifies each crop as object or background



Dog? YES
Cat? NO
Background? NO

Object Detection: Multiple Objects

Apply a CNN to many different crops of the image, CNN classifies each crop as object or background



Dog? NO
Cat? YES
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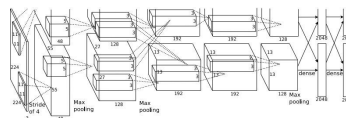
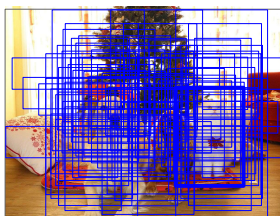


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Object Detection: Multiple Objects

Apply a CNN to many different crops of the image, CNN classifies each crop as object or background



Dog? NO
Cat? YES
Background? NO

Problem: Need to apply CNN to huge number of locations, scales, and aspect ratios, very computationally expensive!



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Object Detection

Two Stages

- Propose “objects”
- Classify each candidate

One-Stage

- Sliding window to
classify all candidates



Object Detection

Two Stages

- Propose “objects”
- Classify each candidate

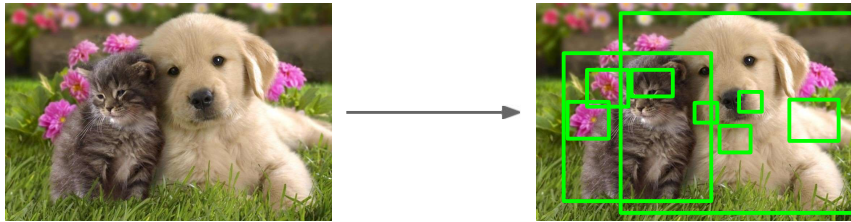
One-Stage

- Sliding window to
classify all candidates



Region Proposals: Selective Search

- Find “blobby” image regions that are likely to contain objects
- Relatively fast to run; e.g. Selective Search gives 2000 region proposals in a few seconds on CPU



Alexe et al, “Measuring the objectness of image windows”, TPAMI 2012 Uijlings et al, “Selective Search for Object Recognition”, IJCV 2013
 Cheng et al, “BING: Binarized normed gradients for objectness estimation at 300fps”, CVPR 2014 Zitnick and Dollar, “Edge boxes: Locating object proposals from edges”, ECCV 2014



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R-CNN



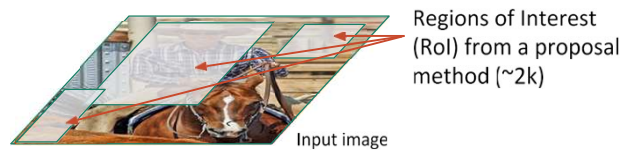
Girshick et al, “Rich feature hierarchies for accurate object detection and semantic segmentation”, CVPR 2014.



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R-CNN



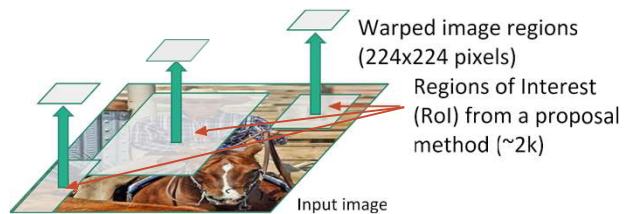
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R-CNN



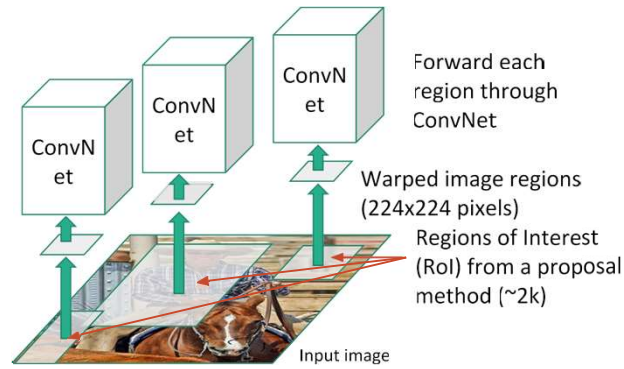
Girshick et al, "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014.



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R-CNN



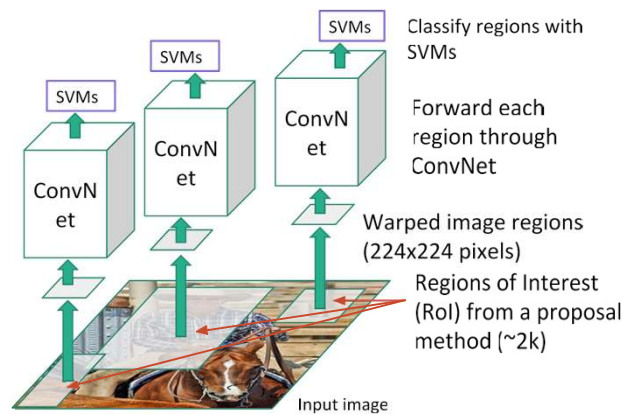
Girshick et al., "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014.



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R-CNN



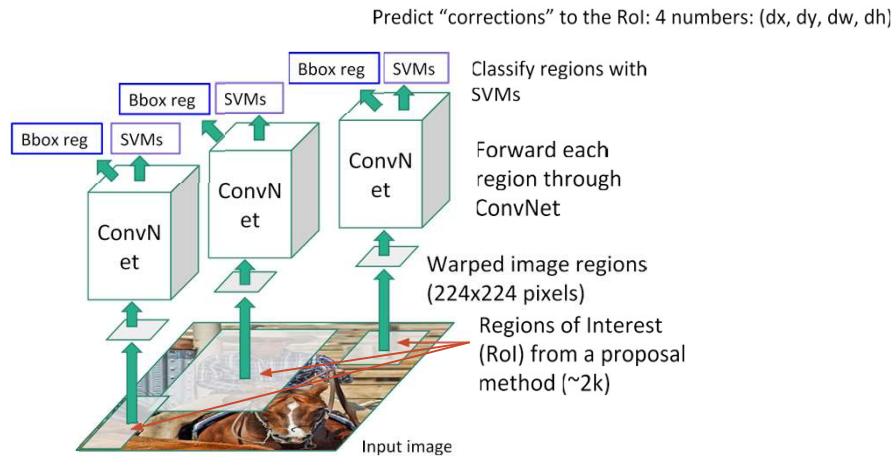
Girshick et al., "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014.



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R-CNN



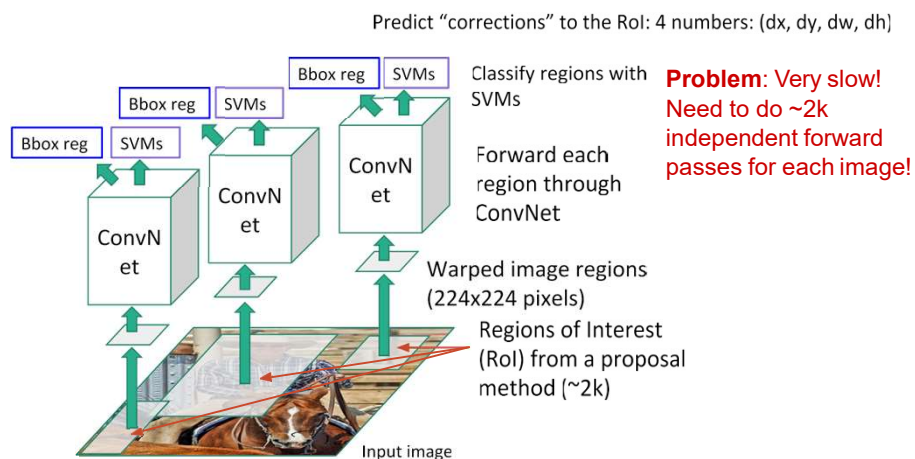
Girshick et al., "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014.



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R-CNN



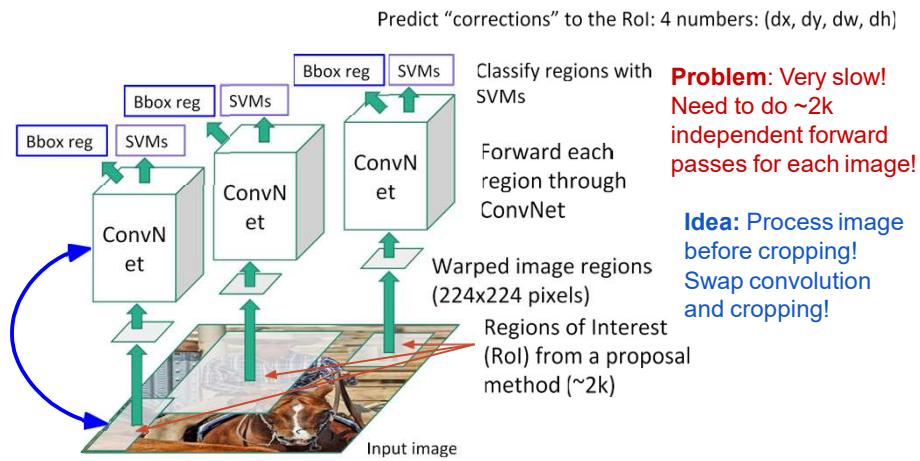
Girshick et al., "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014.



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“Slow” R-CNN



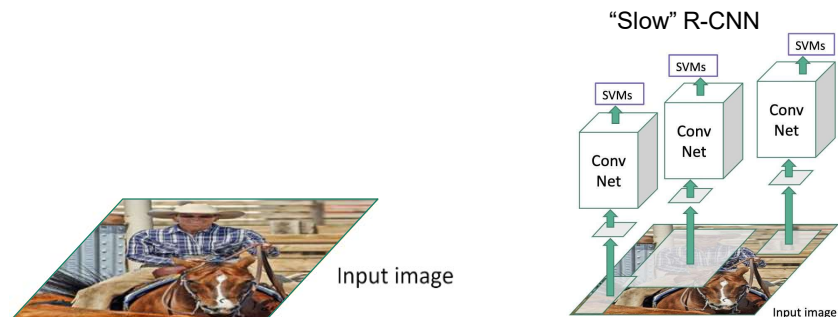
Girshick et al, “Rich feature hierarchies for accurate object detection and semantic segmentation”, CVPR 2014.



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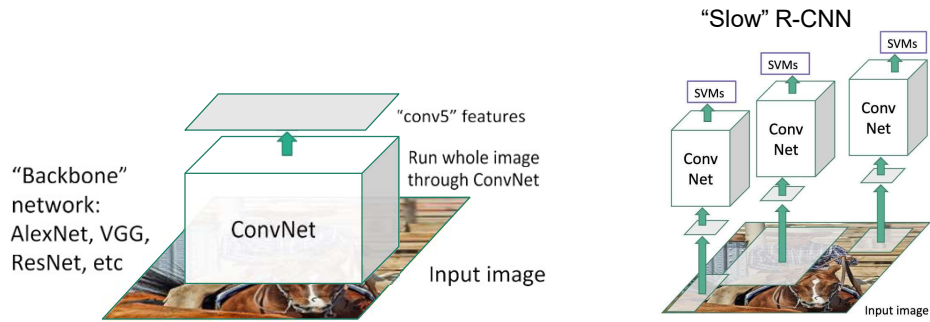
Fast R-CNN



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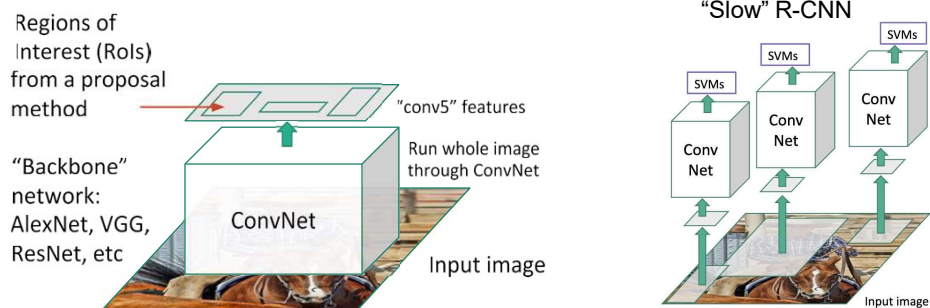
Fast R-CNN



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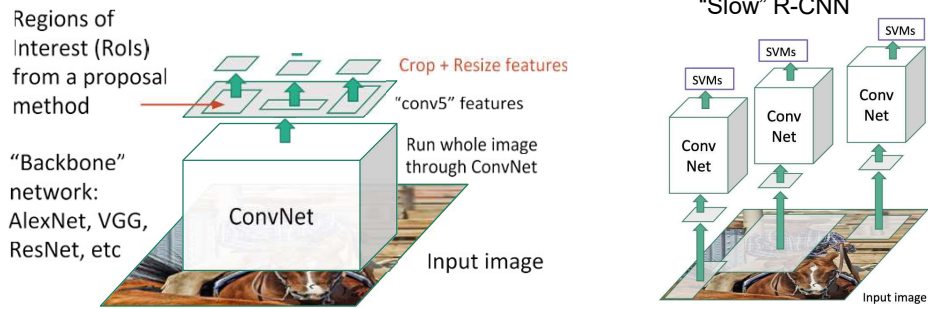
Fast R-CNN



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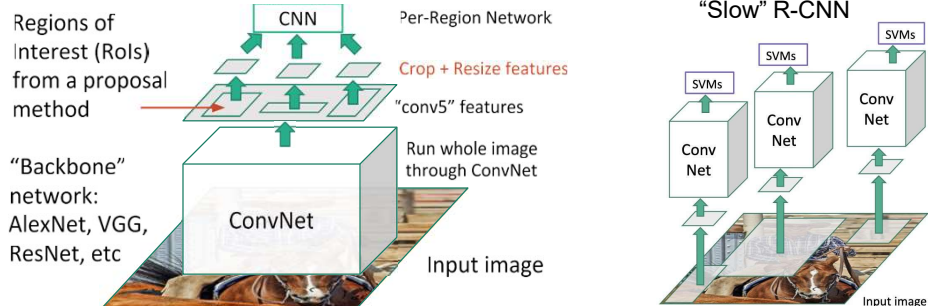
Fast R-CNN



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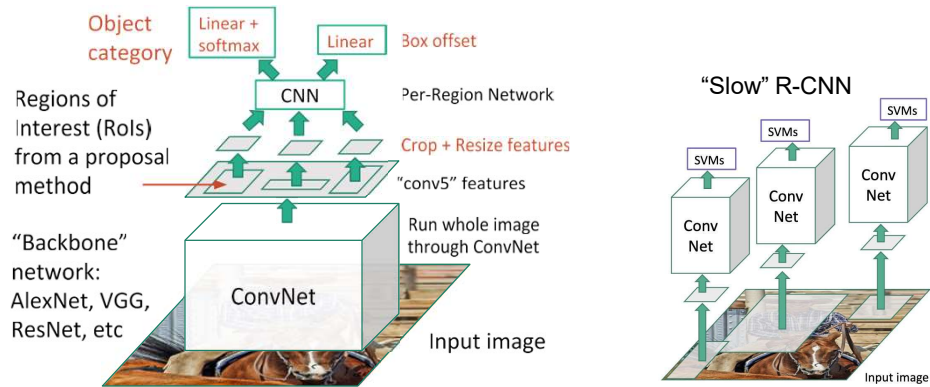
Fast R-CNN



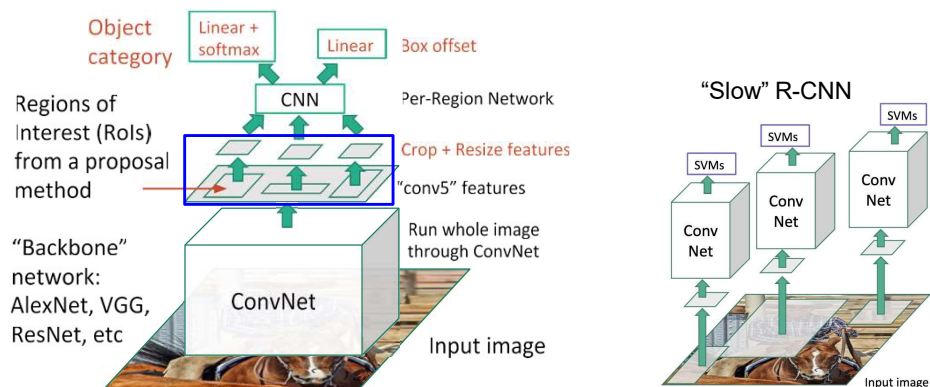
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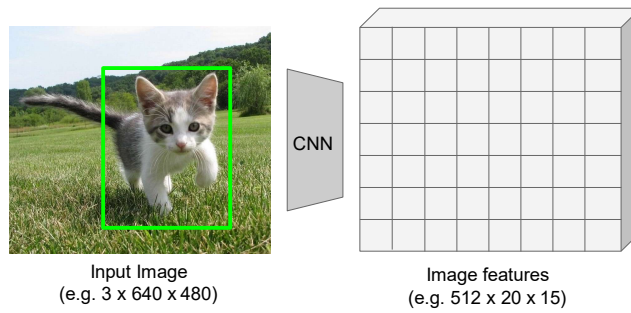
Fast R-CNN



Fast R-CNN



Cropping Features: RoI Pool



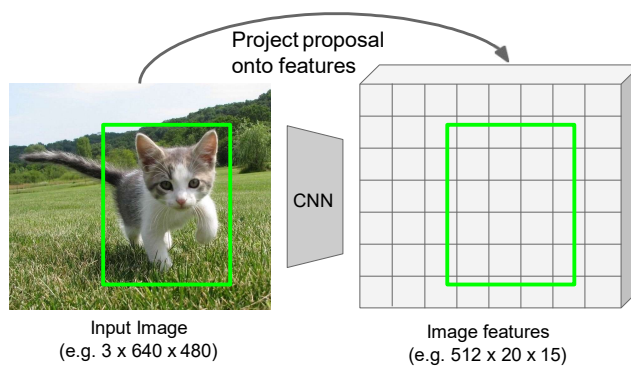
Girshick, "Fast R-CNN", ICCV 2015.



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Cropping Features: RoI Pool



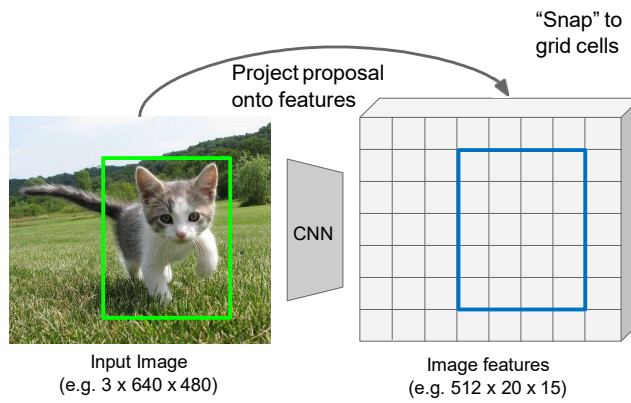
Girshick, "Fast R-CNN", ICCV 2015.



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Cropping Features: RoI Pool



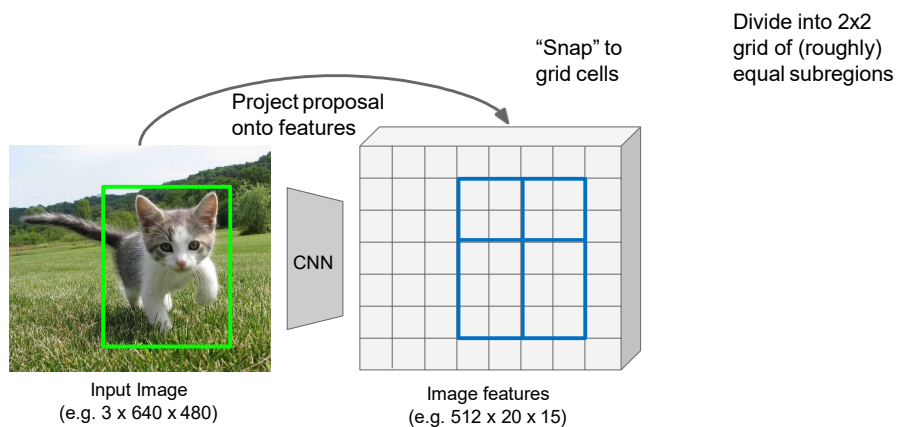
Girshick, "Fast R-CNN", ICCV 2015.



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Cropping Features: RoI Pool



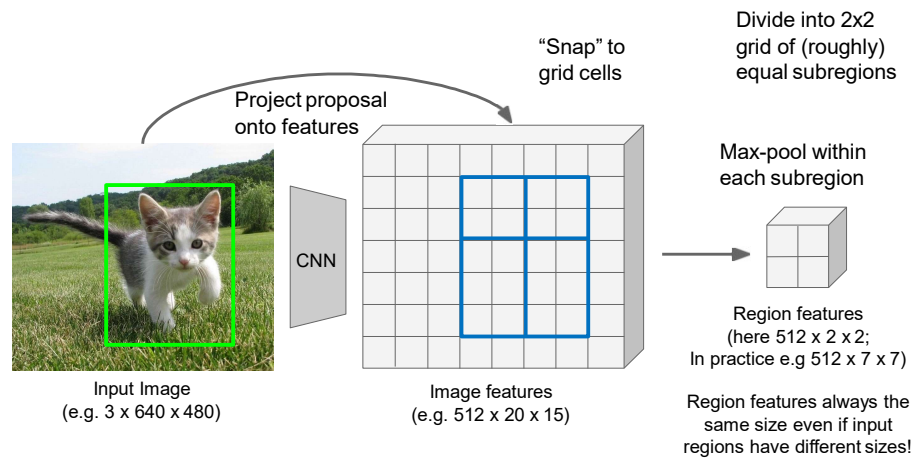
Girshick, "Fast R-CNN", ICCV 2015.



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Cropping Features: RoI Pool



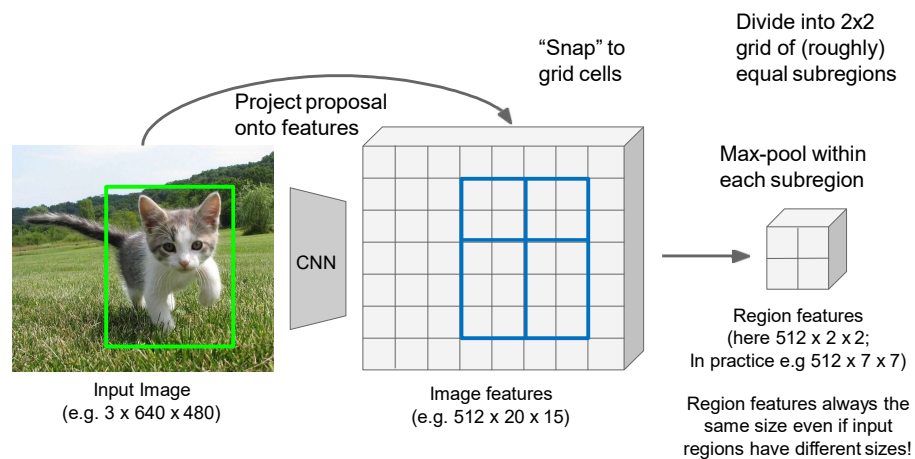
Girshick, "Fast R-CNN", ICCV 2015.



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Cropping Features: RoI Pool



Girshick, "Fast R-CNN", ICCV 2015.

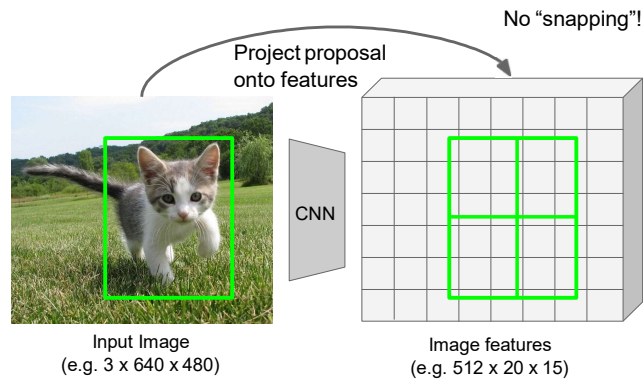
Problem: Region features slightly misaligned



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Cropping Features: RoI Align



Girshick, "Fast R-CNN", ICCV 2015.

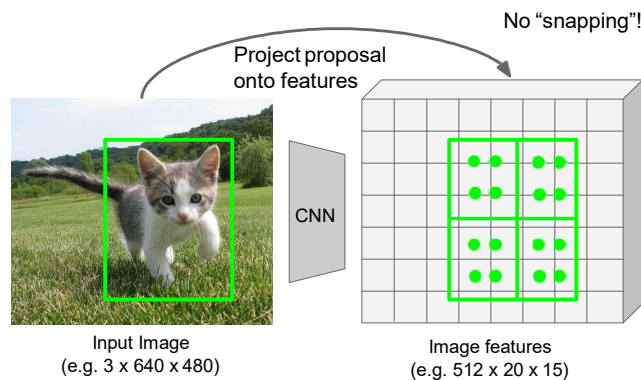


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Cropping Features: RoI Align

Sample at regular points in each subregion using bilinear interpolation



Girshick, "Fast R-CNN", ICCV 2015.

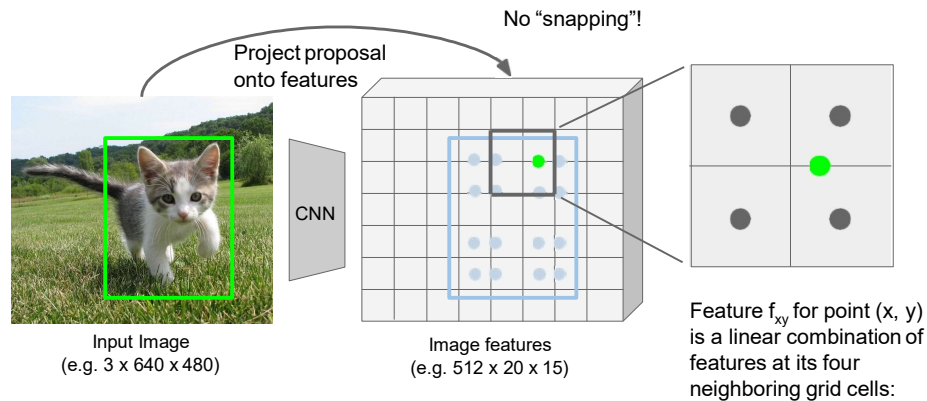


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Cropping Features: RoI Align

Sample at regular points in each subregion using bilinear interpolation



Girshick, "Fast R-CNN", ICCV 2015.

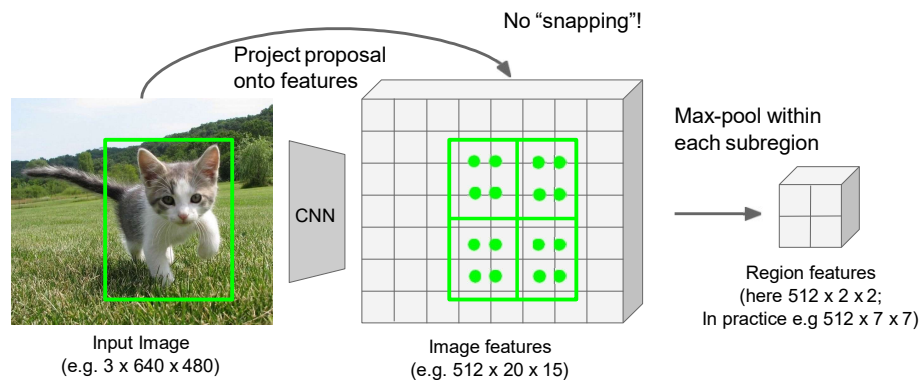


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Cropping Features: RoI Align

Sample at regular points in each subregion using bilinear interpolation



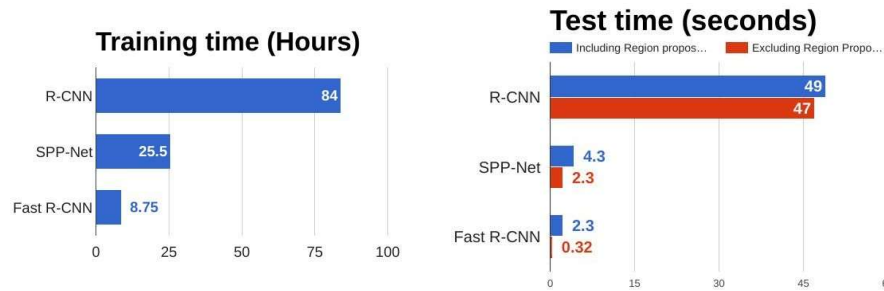
Girshick, "Fast R-CNN", ICCV 2015.



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R-CNN vs Fast R-CNN



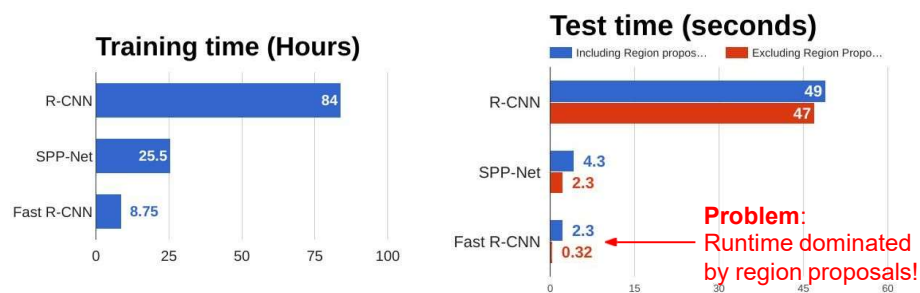
Girshick et al., "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014.
 He et al., "Spatial pyramid pooling in deep convolutional networks for visual recognition", ECCV 2014
 Girshick, "Fast R-CNN", ICCV 2015



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R-CNN vs Fast R-CNN



Girshick et al., "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014.
 He et al., "Spatial pyramid pooling in deep convolutional networks for visual recognition", ECCV 2014
 Girshick, "Fast R-CNN", ICCV 2015



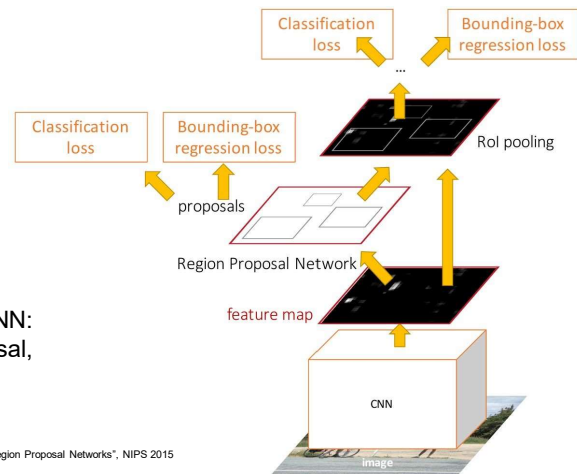
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Faster R-CNN: Make CNN do proposals!

Insert **Region Proposal Network (RPN)** to predict proposals from features

Otherwise same as Fast R-CNN:
Crop features for each proposal,
classify each one



Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NIPS 2015
Figure copyright 2015, Ross Girshick; reproduced with permission



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Region Proposal Network



Input Image
(e.g. 3 x 640 x 480)

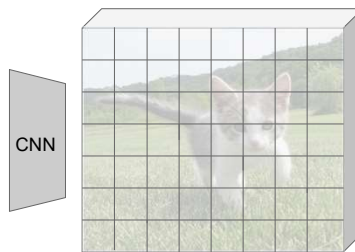


Image features
(e.g. 512 x 20 x 15)



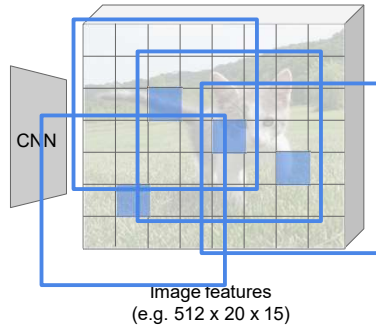
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Region Proposal Network



Input Image
(e.g. 3 x 640 x 480)



Imagine an **anchor box**
of fixed size at each
point in the feature map



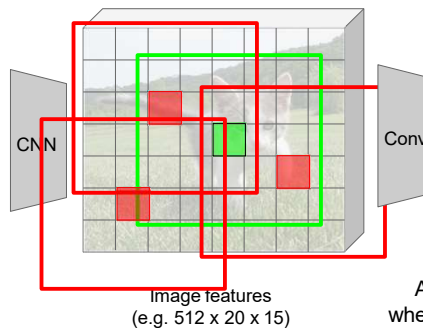
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Region Proposal Network



Input Image
(e.g. 3 x 640 x 480)



Imagine an **anchor box**
of fixed size at each
point in the feature map

Anchor is an object?
1 x 20 x 15

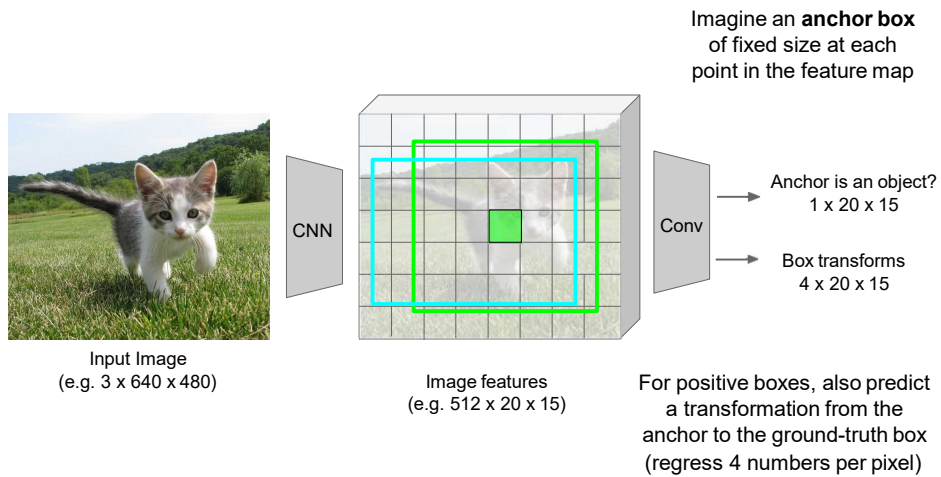
At each point, predict
whether the corresponding
anchor contains an object
(per-pixel logistic regression)



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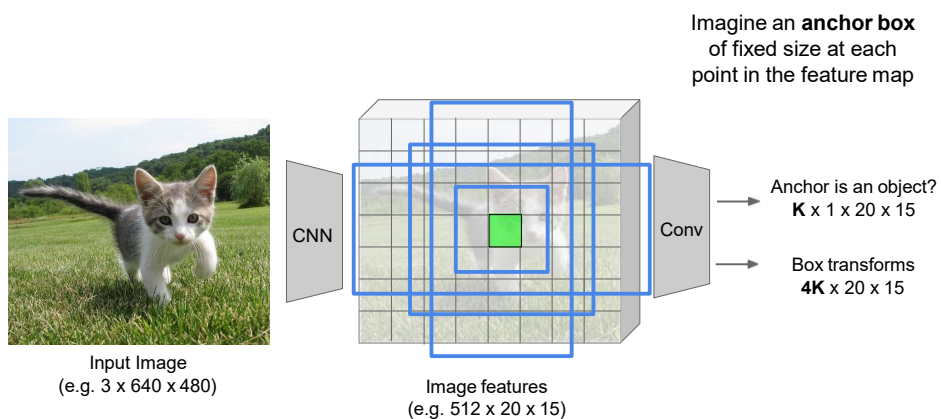
Region Proposal Network



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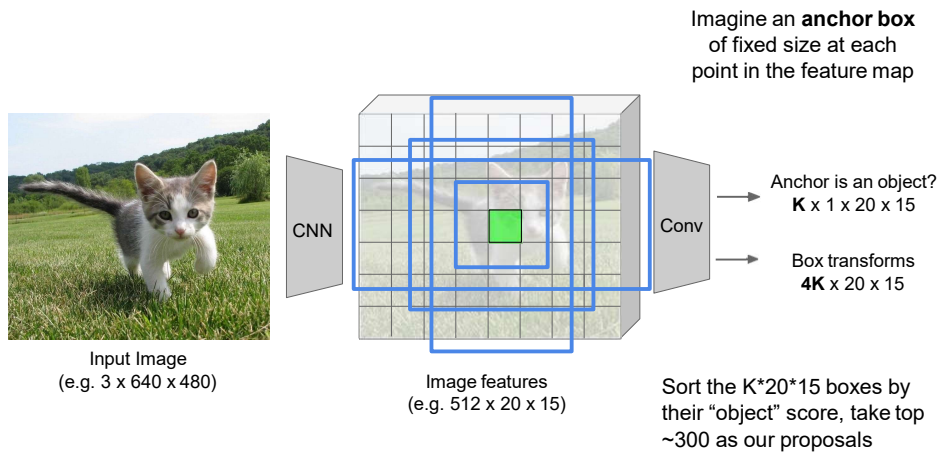
Region Proposal Network



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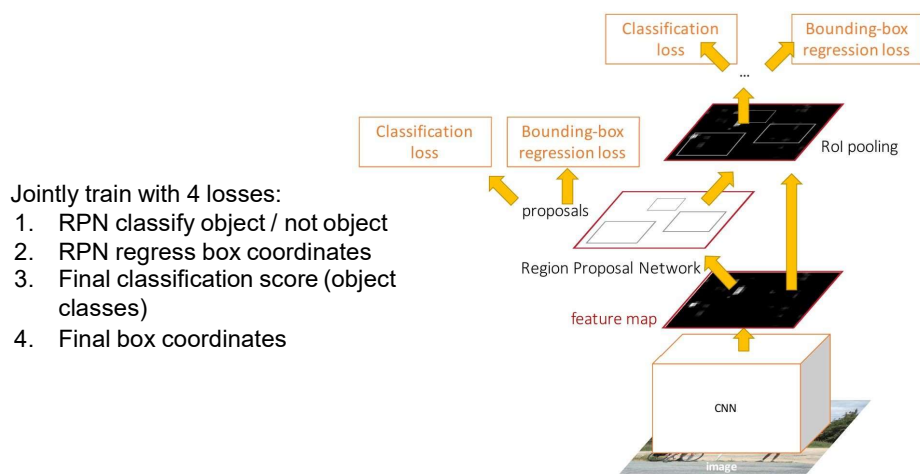
Region Proposal Network



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Faster R-CNN: Make CNN do proposals!



Jointly train with 4 losses:

1. RPN classify object / not object
2. RPN regress box coordinates
3. Final classification score (object classes)
4. Final box coordinates

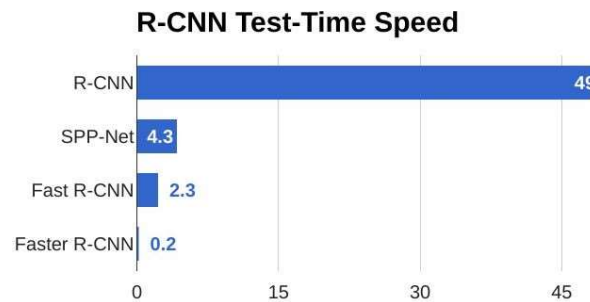
Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NIPS 2015



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Faster R-CNN: Make CNN do proposals!



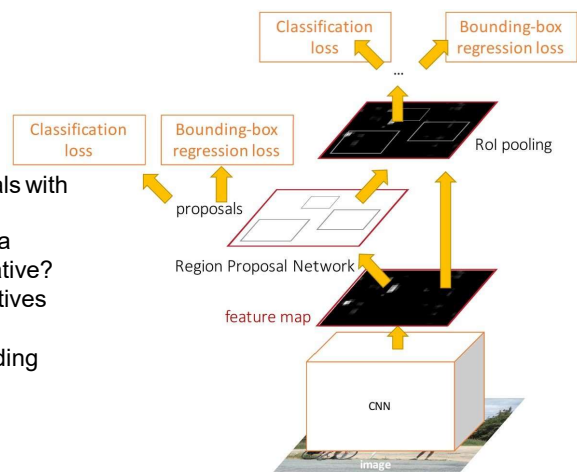
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Faster R-CNN: Make CNN do proposals!

Glossing over many details:

- Ignore overlapping proposals with **non-max suppression**
- How to determine whether a proposal is positive or negative?
- How many positives / negatives to send to second stage?
- How to parameterize bounding box regression?



Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NIPS 2015



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Faster R-CNN: Make CNN do proposals!

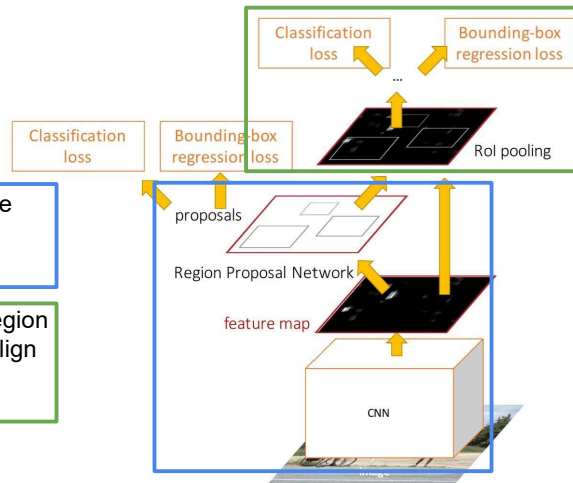
Faster R-CNN is a
Two-stage object detector

First stage: Run once per image

- Backbone network
- Region proposal network

Second stage: Run once per region

- Crop features: RoI pool / align
- Predict object class
- Prediction bbox offset



Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NIPS 2015



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Faster R-CNN: Make CNN do proposals!

Do we really need
the second stage?

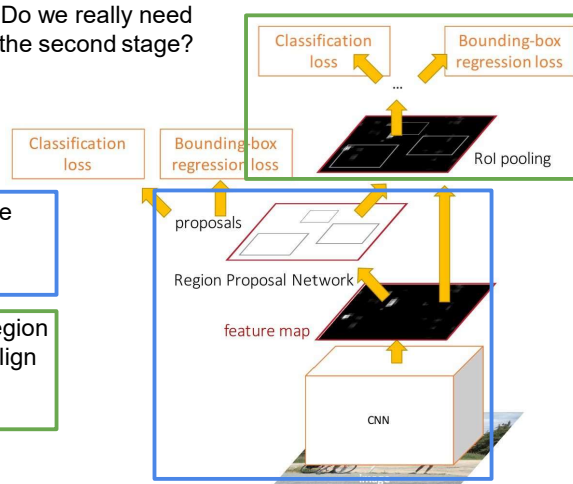
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Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NIPS 2015



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One-stage Object detection

Anchor-based



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Object Detection

Two Stages

- Propose “objects”
- Classify each candidate

One-Stage

- Sliding window to
classify all candidates



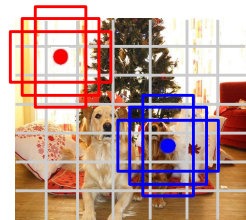
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Single-Stage Object Detectors: YOLO / SSD / RetinaNet



Input image
3 x H x W



Divide image into grid
7 x 7
Image a set of **base boxes**
centered at each grid cell
Here B = 3

Within each grid cell:

- Regress from each of the B base boxes to a final box with 5 numbers: (dx, dy, dh, dw, confidence)
- Predict scores for each of C classes (including background as a class)
- Looks a lot like RPN, but category-specific!

Output:
7 x 7 x (5 * B + C)

Redmon et al, "You Only Look Once: Unified, Real-Time Object Detection", CVPR 2016
Liu et al, "SSD: Single-Shot MultiBox Detector", ECCV 2016
Lin et al, "Focal Loss for Dense Object Detection", ICCV 2017



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Imbalance

Number of "negative" anchors $\sim O(10K)$

Number of "positive" anchors $\sim O(10)$

What happens to CE loss in this case?

$$\mathcal{L} = -\frac{1}{N} \sum_i y_i \log(p_i), \quad \frac{\partial \mathcal{L}}{\partial p_i} = \begin{cases} p_{i,l} & \text{if } y_i \neq l \\ p_{i,l} - 1 & \text{if } y_i = l \end{cases}$$

Loss and gradient are dominated by correctly classified negative examples

Training outcome $\rightarrow$ constant "negative" prediction.



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Imbalance – Focal Loss

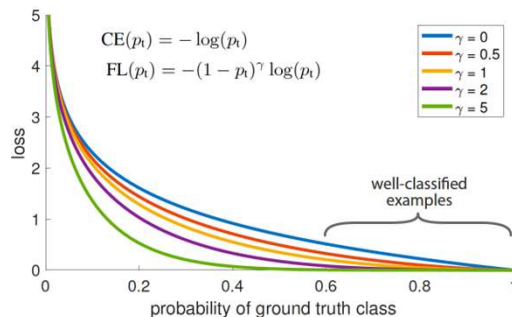


Figure 1: We propose a novel loss we term the *Focal Loss* that adds a factor $(1 - p_t)^\gamma$ to the standard cross entropy criterion. Setting $\gamma > 0$ reduces the relative loss for well-classified examples ($p_t > .5$), putting more focus on hard, misclassified examples. As our experiments will demonstrate, the proposed focal loss enables training highly accurate dense object detectors in the presence of vast numbers of easy background examples.



Lin, Goyal, Girshick, He, and Dollár

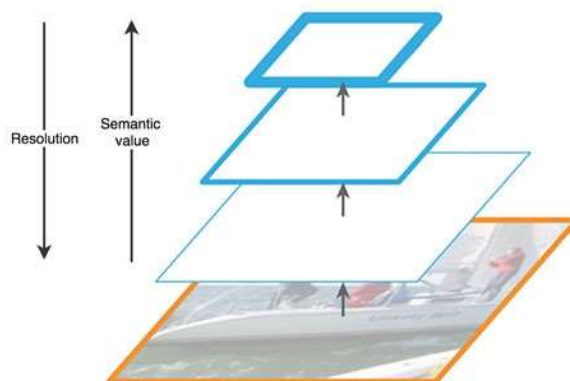
[Focal loss for dense object detection](#) (PAMI 2018)

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Feature Pyramid Network (FPN)

- How to handle multiscale predictions?



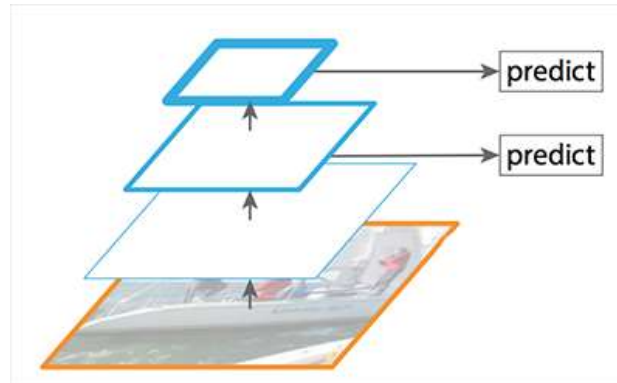
Tsung-Yi, Dollár, Girshick, He, Hariharan and Belongie. [Feature Pyramid Networks for Object Detection](#) (CVPR 2017)

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66

Feature Pyramid Network (FPN)

- How to handle multiscale predictions?

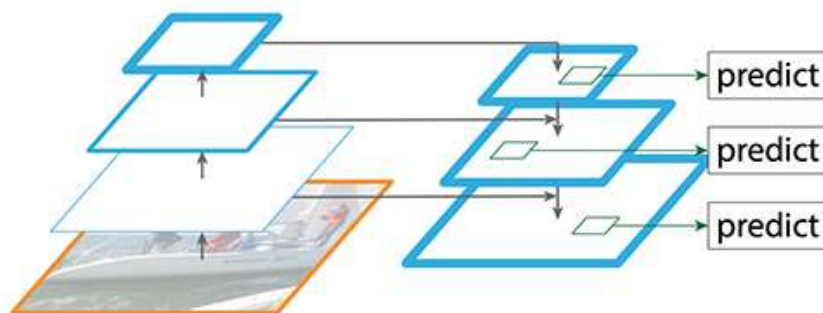


Tsung-Yi, Dollár, Girshick, He, Hariharan and Belongie. [Feature Pyramid Networks for Object Detection](#) (CVPR 2017)
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Feature Pyramid Network (FPN)

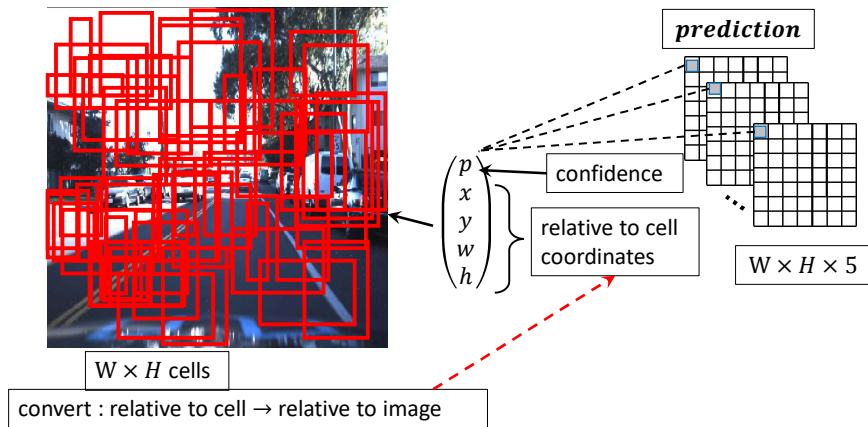
- How to handle multiscale predictions?



Tsung-Yi, Dollár, Girshick, He, Hariharan and Belongie. [Feature Pyramid Networks for Object Detection](#) (CVPR 2017)
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Postprocessing: NMS



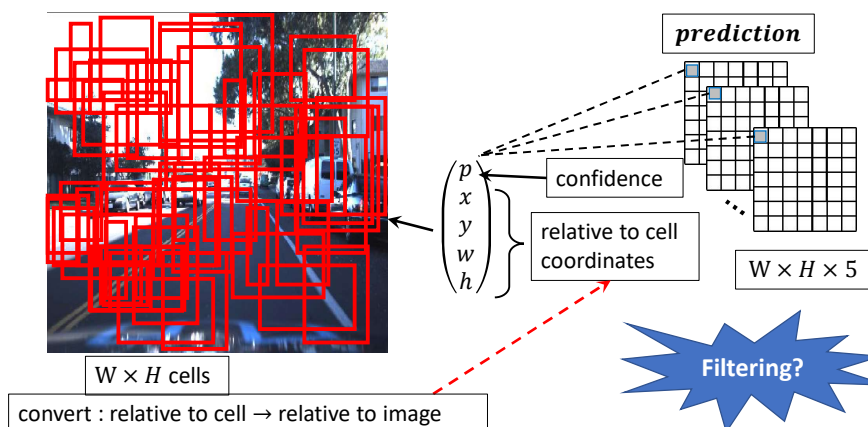
69



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69

Postprocessing: NMS



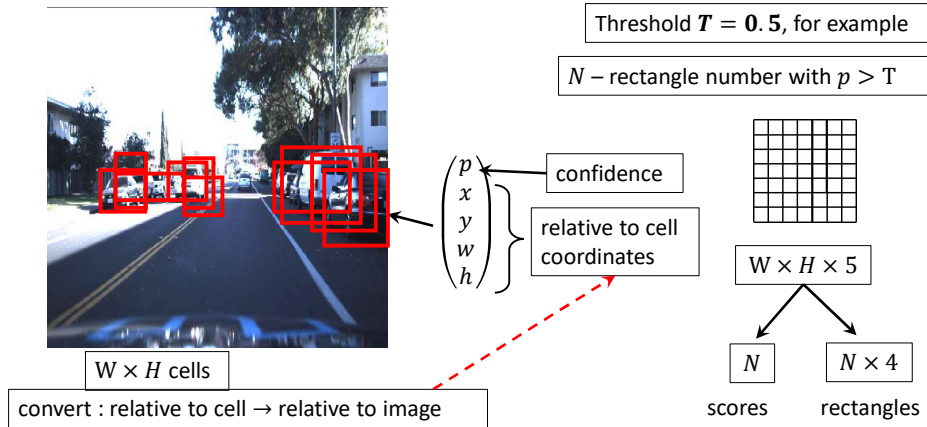
70



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70

Postprocessing: NMS



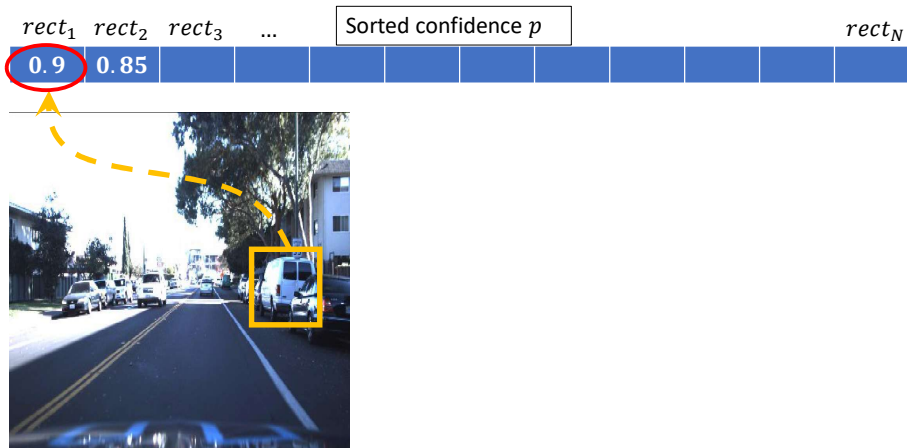
71



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71

Postprocessing: NMS



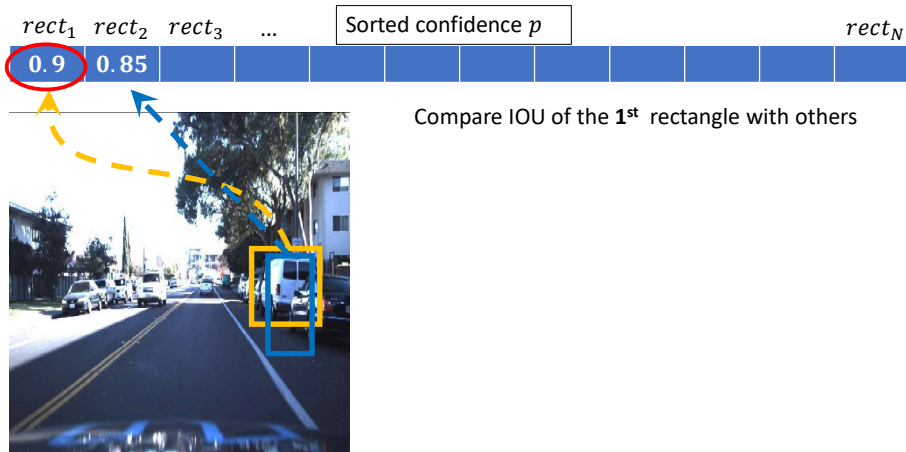
72



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72

Postprocessing: NMS



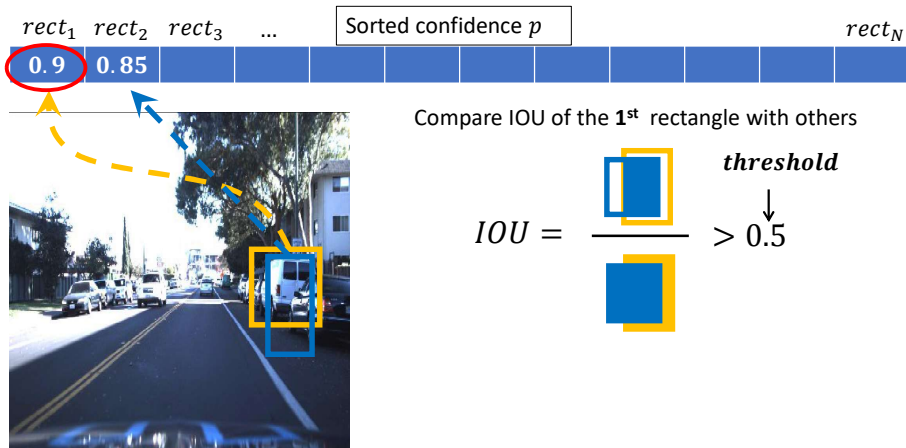
73



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73

Postprocessing: NMS



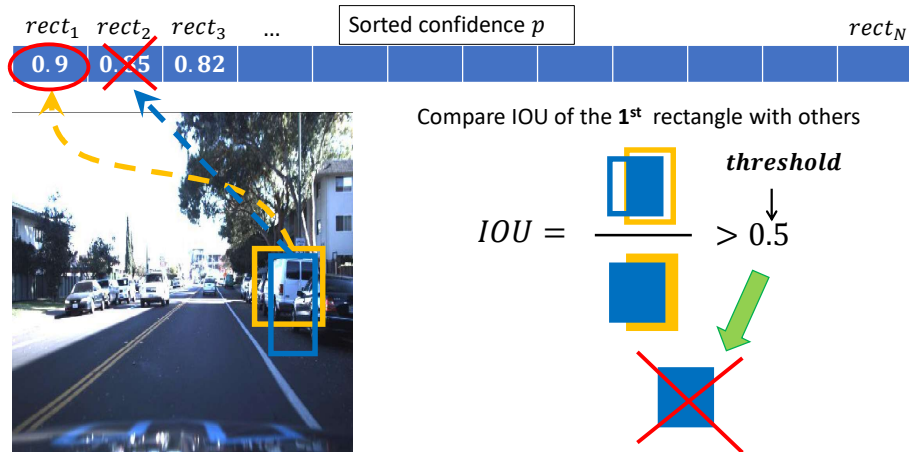
74



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74

Postprocessing: NMS



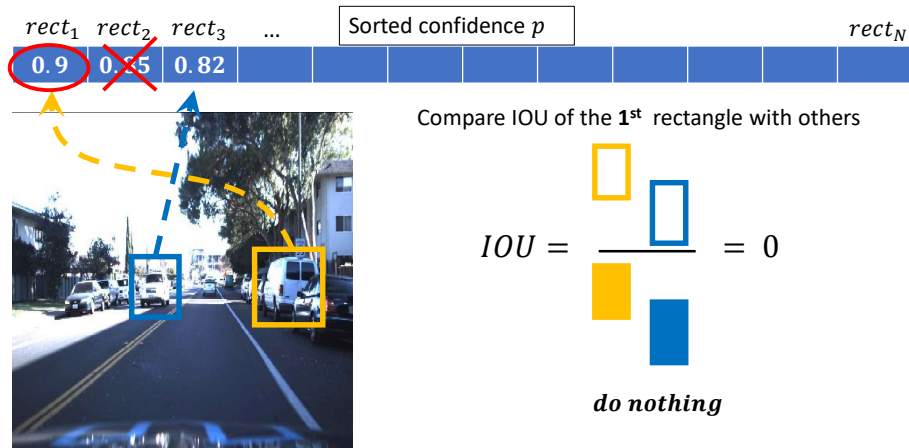
75



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75

Postprocessing: NMS



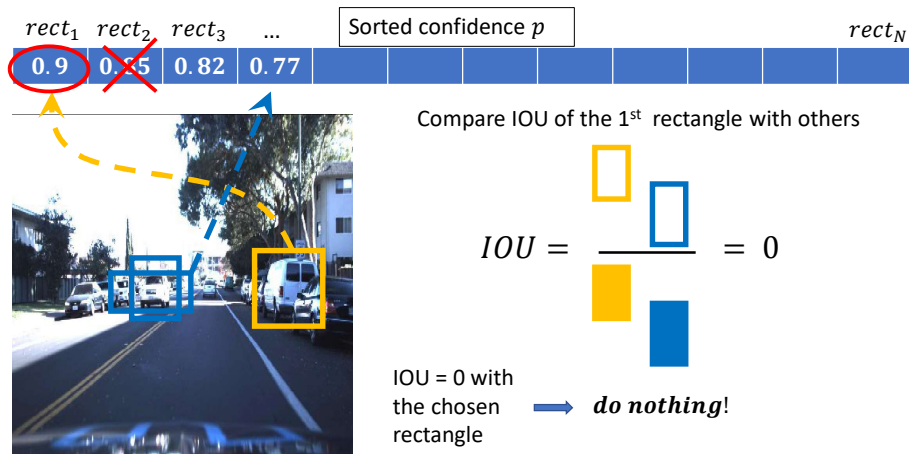
76



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76

Postprocessing: NMS



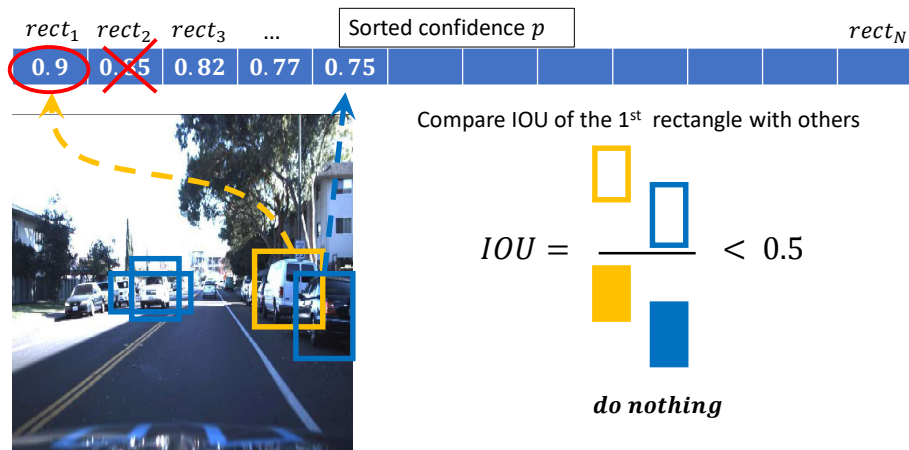
77



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Postprocessing: NMS



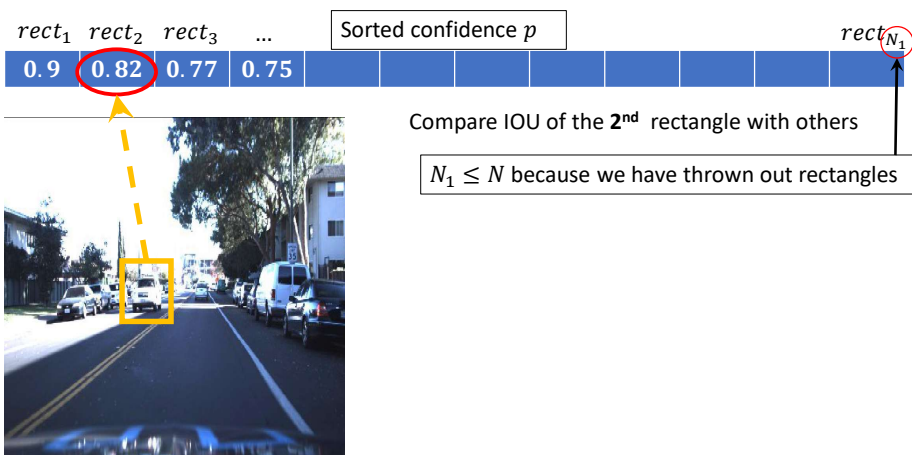
78



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Postprocessing: NMS



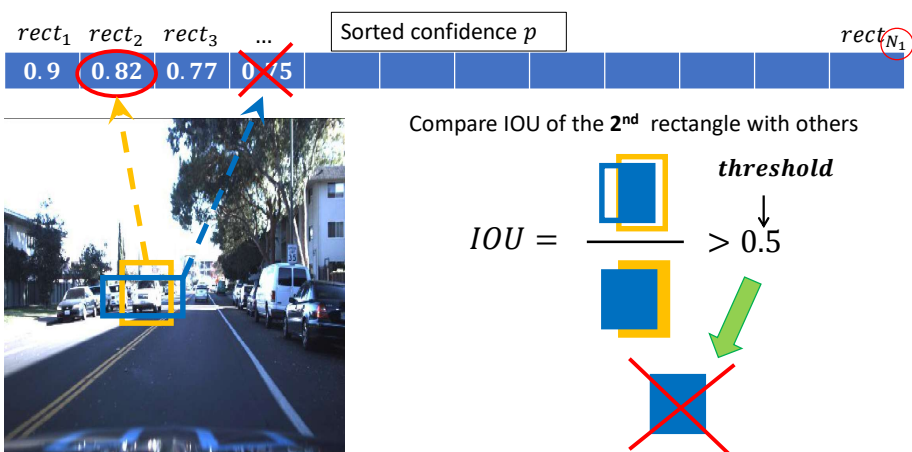
79



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Postprocessing: NMS



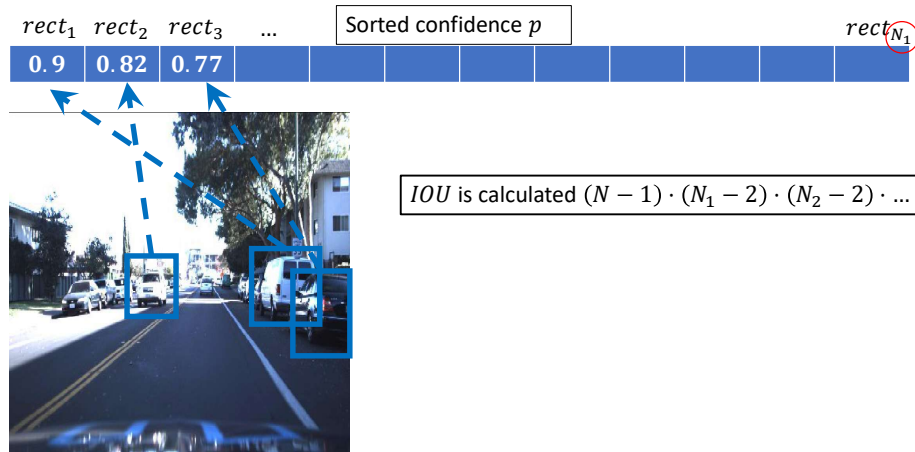
80



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Postprocessing: NMS



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One-stage Object detection

Anchor-free



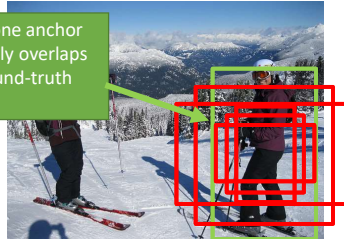
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Drawbacks of Anchor Boxes

1. Need a large number of anchors

At least one anchor sufficiently overlaps with ground-truth



- A tiny fraction of anchors are positive examples
- Slow down training [Lin et al. ICCV'17]

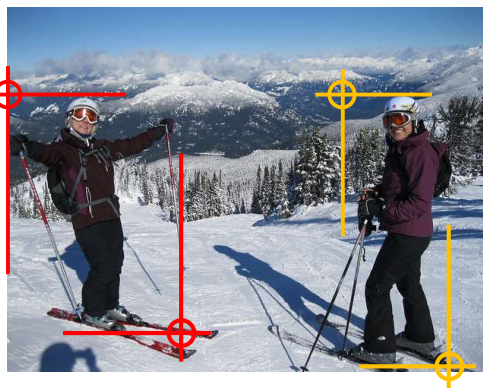
2. Extra hyperparameters – sizes and aspect ratios



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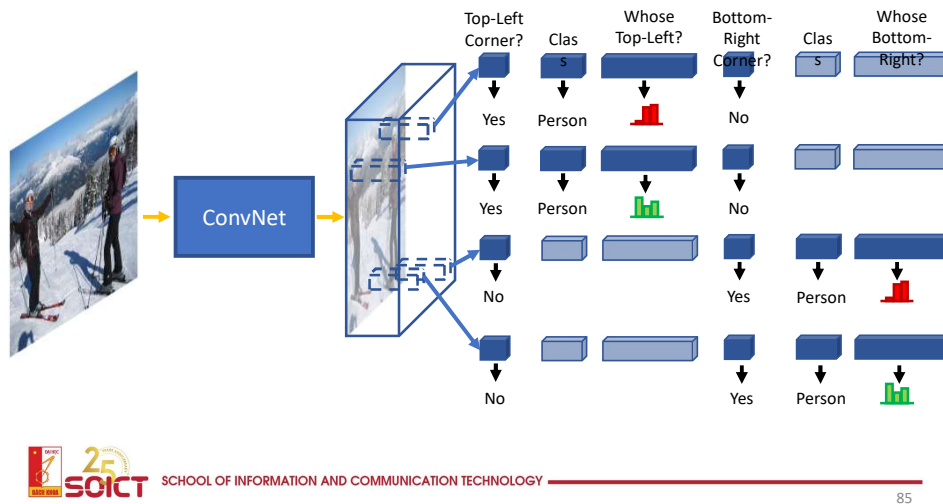
CornerNet: Detecting Objects as Paired Keypoints



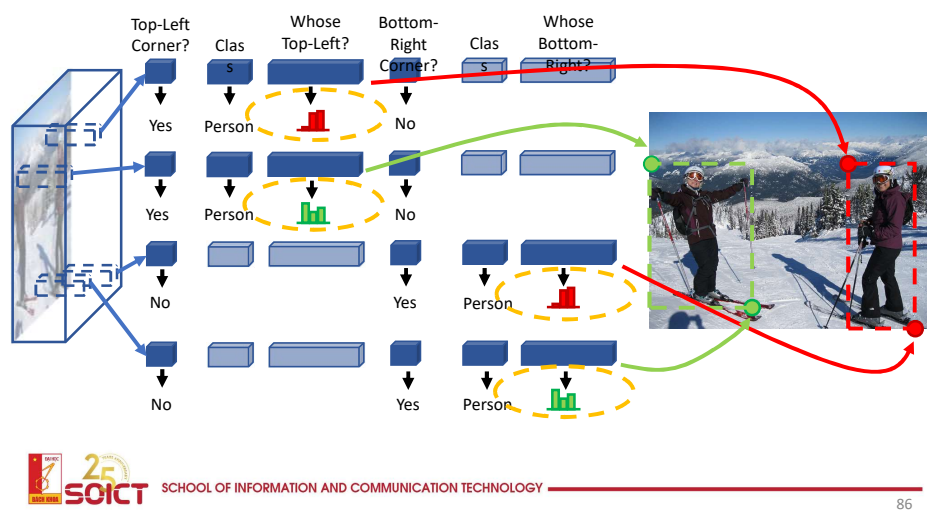
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84

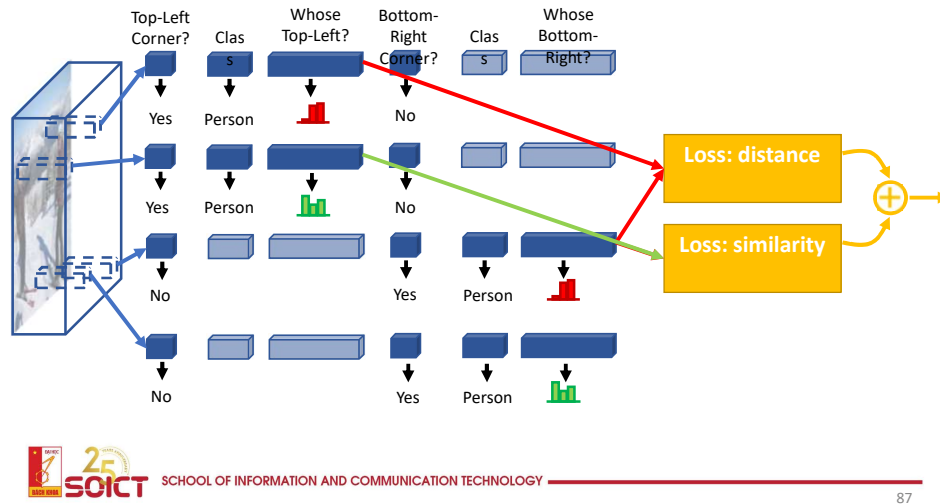
CornerNet: Detecting Objects as Paired Keypoints



CornerNet: Detecting Objects as Paired Keypoints

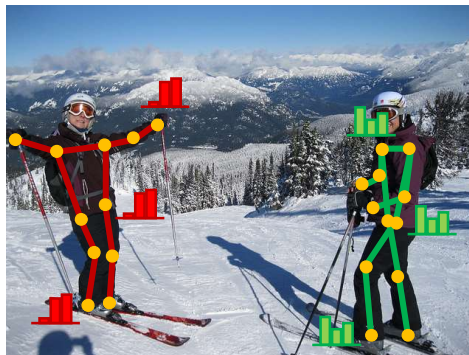


CornerNet: Detecting Objects as Paired Keypoints



Associative Embedding [Newell et al. NIPS'17]

- CornerNet is inspired by the idea of associative embedding from work by Newell et al. [NIPS'17]

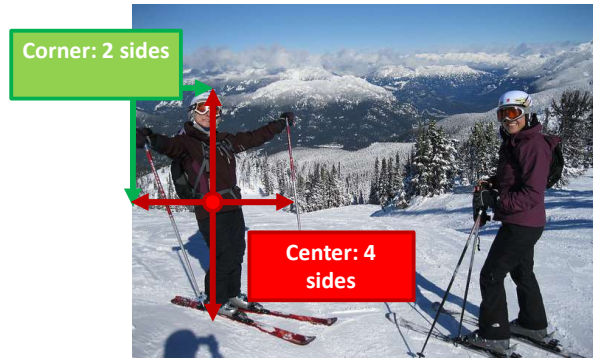


<https://proceedings.neurips.cc/paper/2017/file/8edd72158ccd2a879f79cb2538568fd>

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Advantages of Detecting Corners



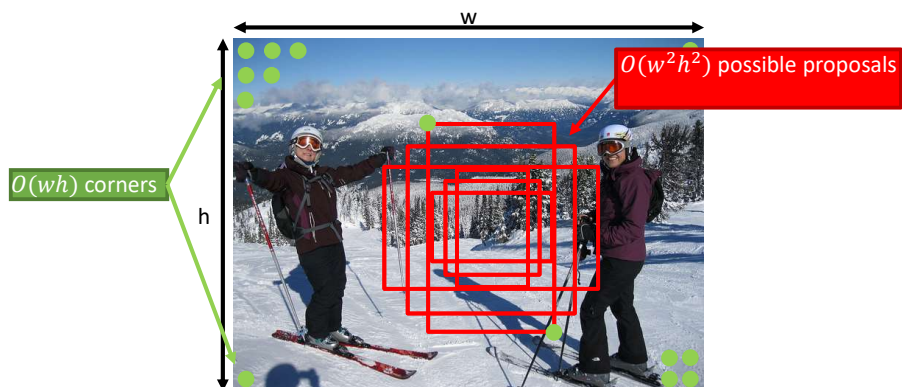
Detecting corner is easier than detecting center



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Advantages of Detecting Corner



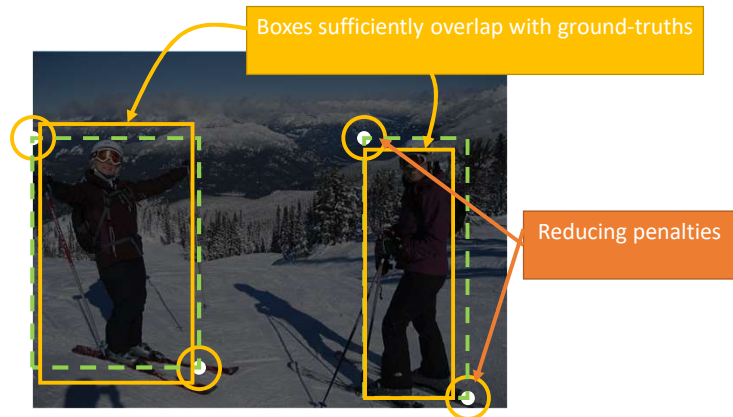
Represent $O(w^2h^2)$ possible proposals using only $O(wh)$ corners



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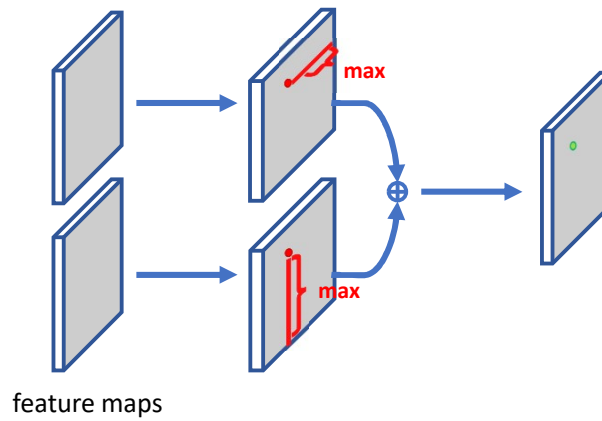
Supervising Corner Detection



Corner Pooling



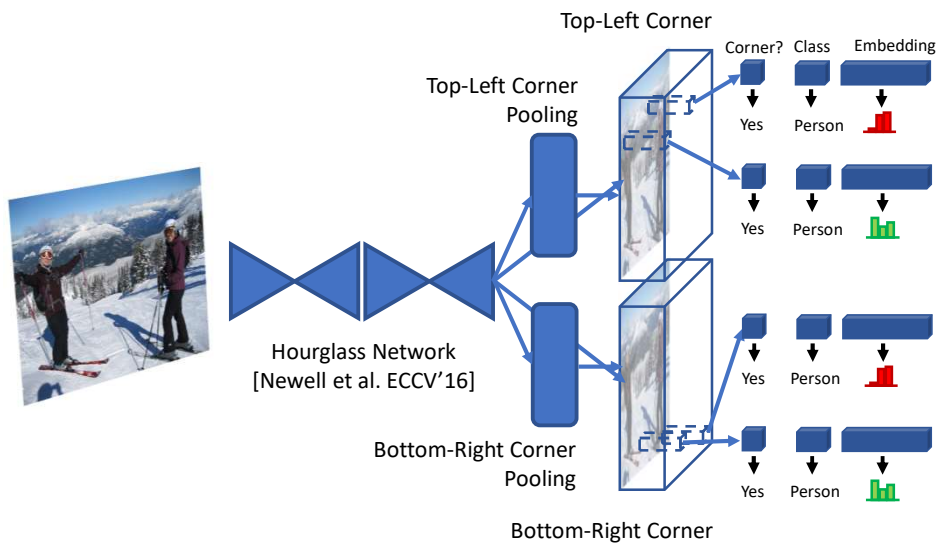
Top-Left Corner Pooling



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CornerNet



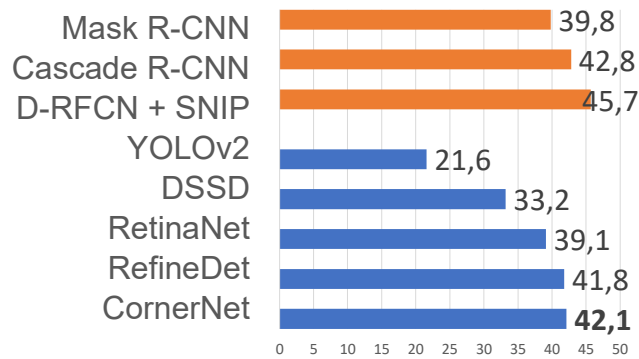
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Experiment: CornerNet versus Others



■ Two-stage ■ One-stage



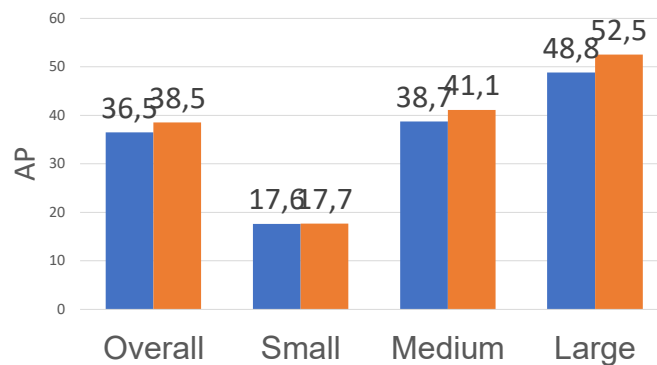
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Experiment: Corner Pooling

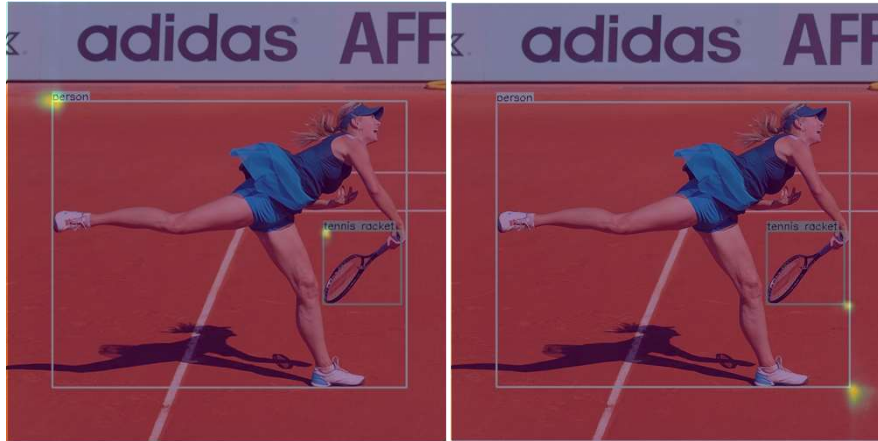


■ w/o corner pooling ■ w/ corner pooling



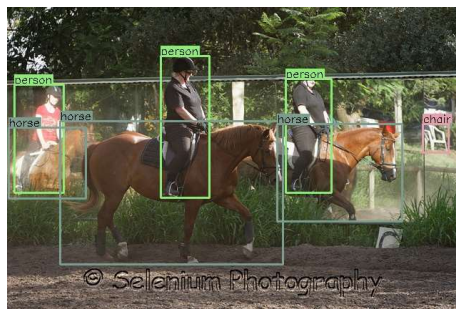
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Accuracy evaluation



 n predicted
 m Ground Truth

$$IoU = \frac{A \cap B}{A \cup B}$$

A – ground truth
 B – detector result

Correspondence between
 GT and predicted?

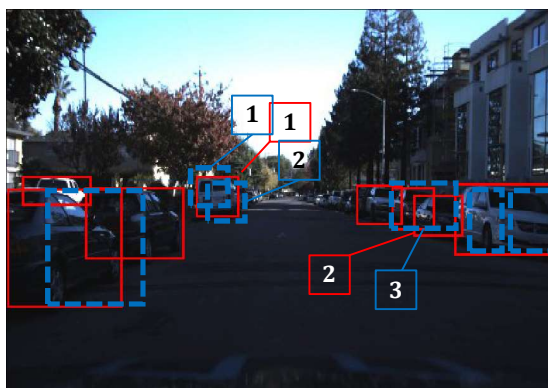
99



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Accuracy evaluation



Sorted array of $IoU > T$
 between GT and predicted
 (max length = $n \cdot m$)

↑
 $(iou_1, iou_2, iou_3, \dots)$
 ↓

GT	2	1	1
pred	3	1	2

evaluator.py -> sort_iou(gt_boxes, pred_boxes, iou_thr)

100



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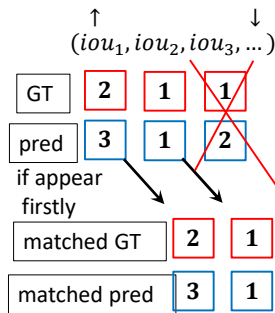
100

Accuracy evaluation



evaluator.py ->
get_single_image_results(gt_boxes, pred_boxes, iou_thr)

Sorted array of $IOU > T$
between GT and predicted
(max length = $n \cdot m$)



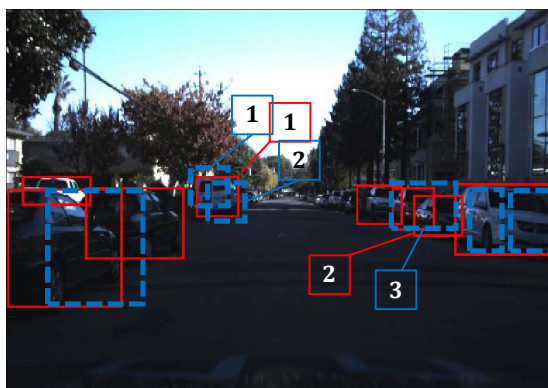
101



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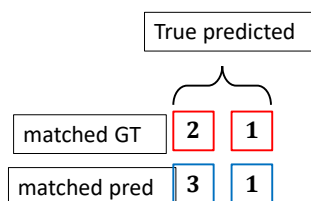
101

Accuracy evaluation



evaluator.py ->
get_single_image_results(gt_boxes, pred_boxes, iou_thr)

Sorted array of $IOU > T$
between GT and predicted
(max length = $n \cdot m$)



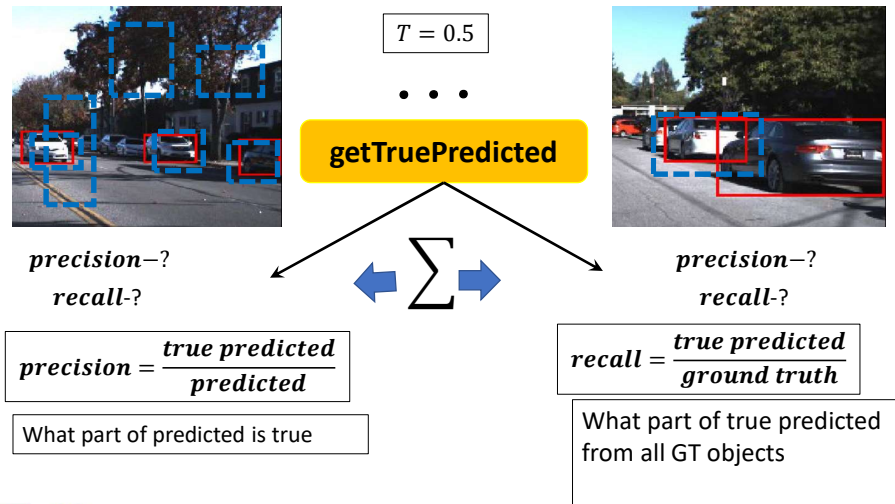
102



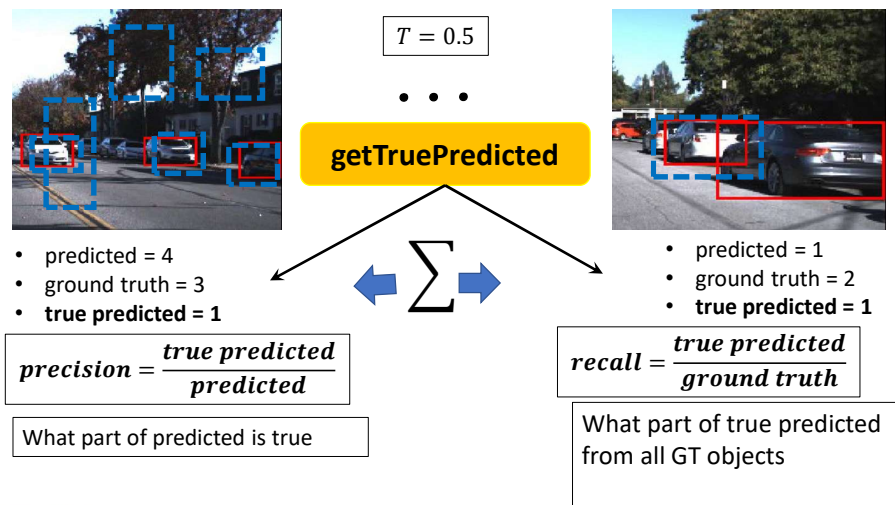
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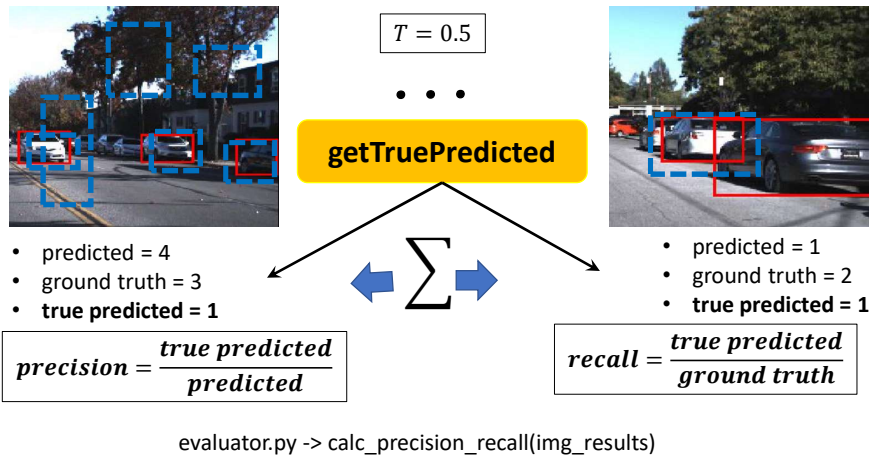
Accuracy evaluation: precision, recall



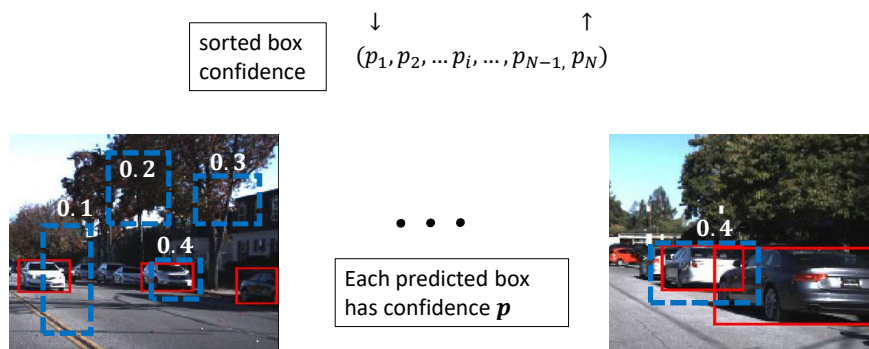
Accuracy evaluation: precision, recall



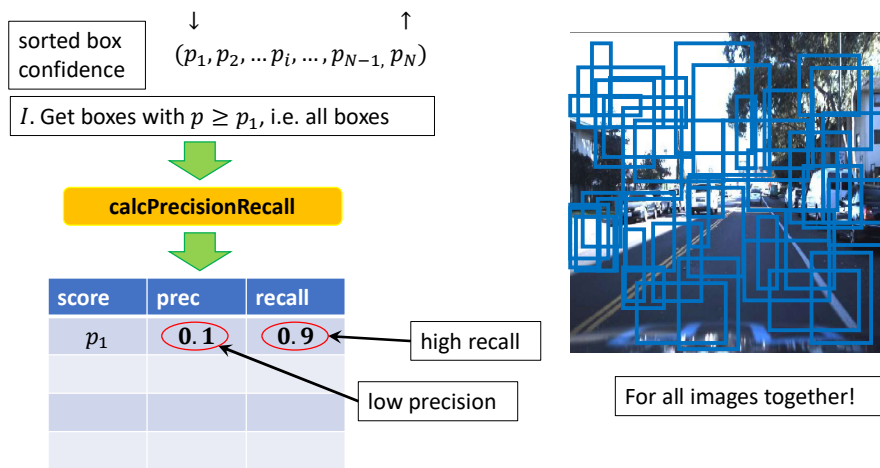
Accuracy evaluation: precision, recall



Accuracy evaluation: precision, recall



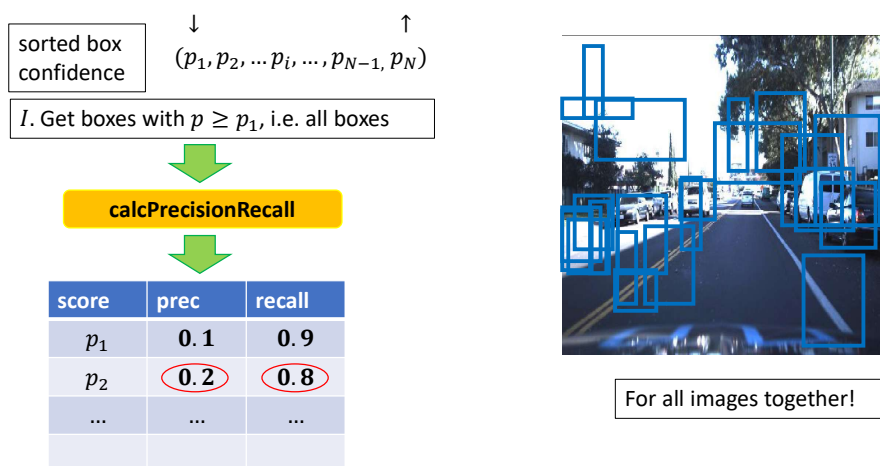
Accuracy evaluation: precision, recall



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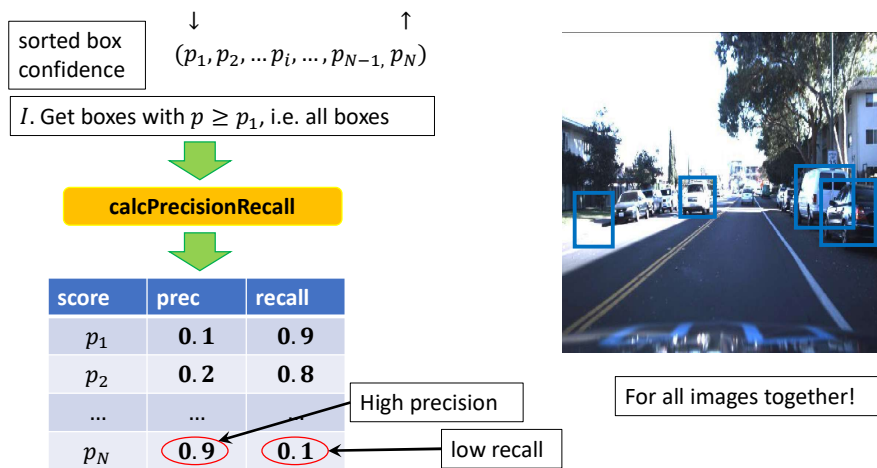
Accuracy evaluation: precision, recall



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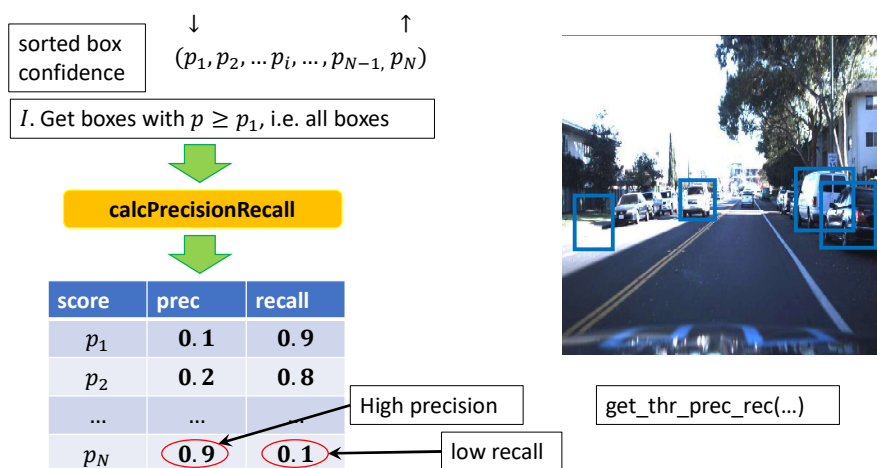
Accuracy evaluation: precision, recall



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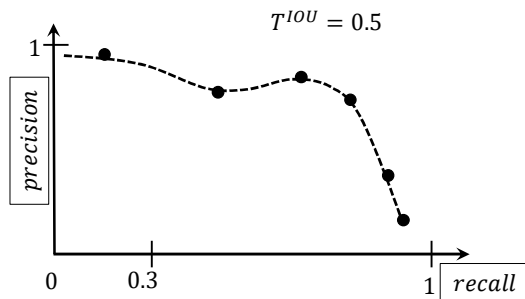
Accuracy evaluation: precision, recall



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Accuracy evaluation: mAP



hundreds of values for
real data sets

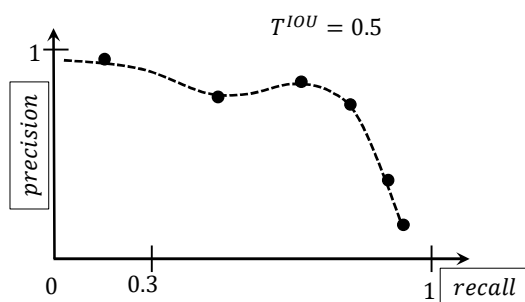
score	prec	recall
p_1	0.1	0.9
p_2	0.2	0.8
...	...	...
p_N	0.9	0.1



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Accuracy evaluation: mAP



hundreds of values for
real data sets

score	prec	recall
p_1	0.1	0.9
p_2	0.2	0.8
...	...	...
p_N	0.9	0.1

11 recall thresholds

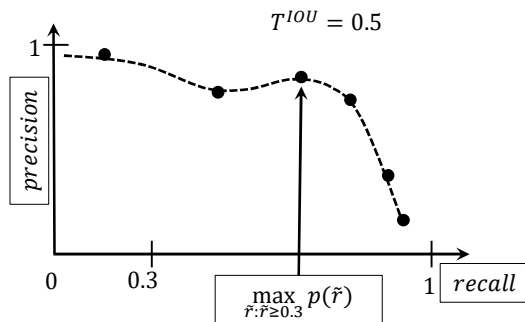
$$AP^T = \sum_{r \in \{0, 0.1, \dots, 1\}} \max_{\tilde{r}: \tilde{r} \geq r} p(\tilde{r})$$



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Accuracy evaluation: mAP



hundreds of values for
real data sets

score	prec	recall
p_1	0.1	0.9
p_2	0.2	0.8
...	...	...
p_N	0.9	0.1

A lot of mAP tutorials has
misunderstanding : “area
under curve” (AUC) is not the
same!

11 recall thresholds

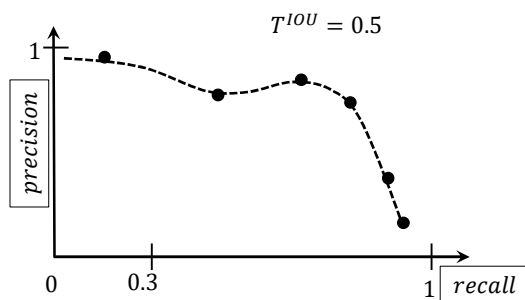
$$AP^T = \sum_{r \in \{0, 0.1, \dots, 1\}} \max_{\tilde{r}: \tilde{r} \geq r} p(\tilde{r})$$



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Accuracy evaluation: mAP



hundreds of values for
real data sets

score	prec	recall
p_1	0.1	0.9
p_2	0.2	0.8
...	...	...
p_N	0.9	0.1

$$AP^T = \sum_{r \in \{0, 0.1, \dots, 1\}} \max_{\tilde{r}: \tilde{r} \geq r} p(\tilde{r})$$

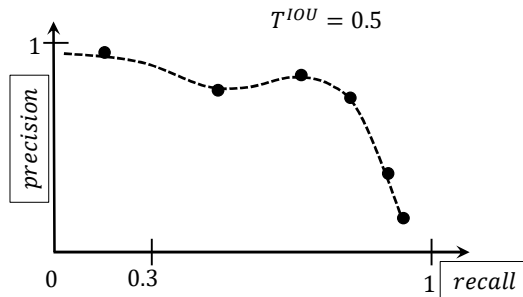
Pascal VOC metric is $AP^{0.5}$



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Accuracy evaluation: mAP



hundreds of values for
real data sets

score	prec	recall
p_1	0.1	0.9
p_2	0.2	0.8
...	...	...
p_N	0.9	0.1

$$AP = \frac{1}{10} \cdot \sum_{T \in \{0.5, 0.55, \dots, 0.95\}} AP^T$$

MS COCO has metrics $AP^{0.5}$, $AP^{0.75}$, AP



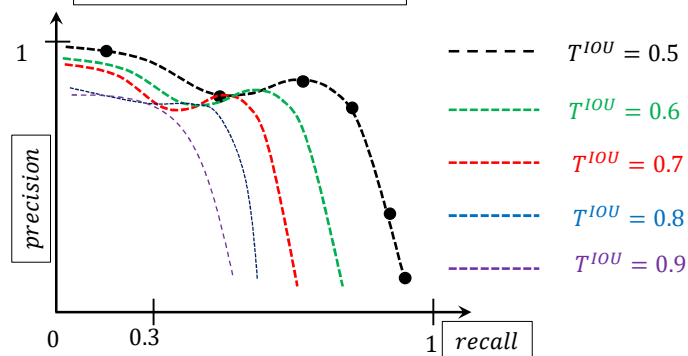
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Accuracy evaluation: mAP

$$AP = \frac{1}{10} \cdot \sum_{T \in \{0.5, 0.55, \dots, 0.95\}} AP^T$$

MS COCO uses 10 IOU
thresholds 0.5, 0.55, ... 0.95



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Computer Vision Tasks

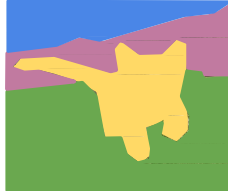
Classification



CAT

No spatial extent

Semantic Segmentation



**GRASS, CAT,
TREE, SKY**

No objects, just pixels

Object Detection



DOG, DOG, CAT

Instance Segmentation



DOG, DOG, CAT

Multiple Object



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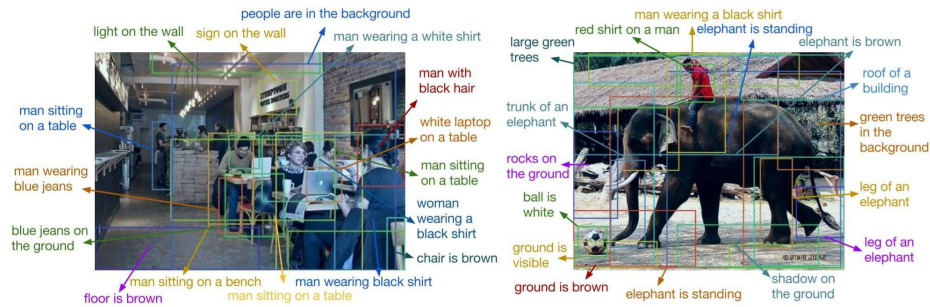
Beyond 2D Object Detection...



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Object Detection + Captioning = Dense Captioning



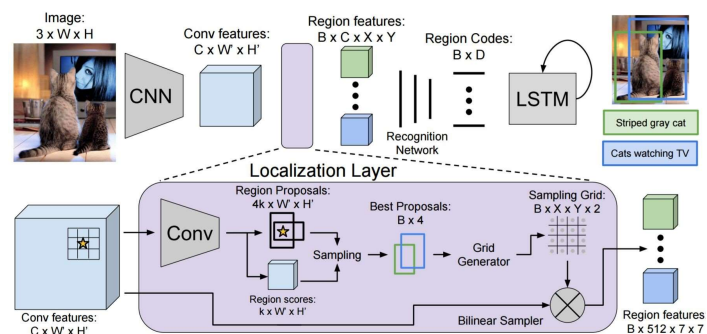
Johnson, Karpathy, and Fei-Fei, "DenseCap: Fully Convolutional Localization Networks for Dense Captioning", CVPR 2016
Figure copyright IEEE, 2016. Reproduced for educational purposes.



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Aside: Object Detection + Captioning = Dense Captioning



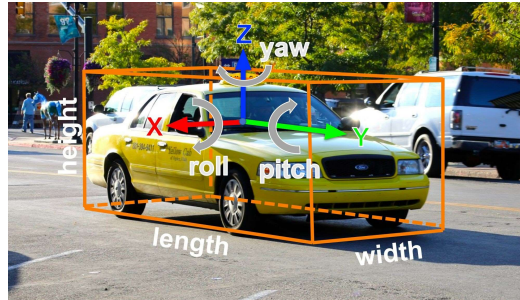
Johnson, Karpathy, and Fei-Fei, "DenseCap: Fully Convolutional Localization Networks for Dense Captioning", CVPR 2016
Figure copyright IEEE, 2016. Reproduced for educational purposes.



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3D Object Detection



2D Object Detection:

2D bounding box
(x, y, w, h)

3D Object Detection:

3D oriented bounding box
(x, y, z, w, h, l, r, p, y)

Simplified bbox: no roll & pitch

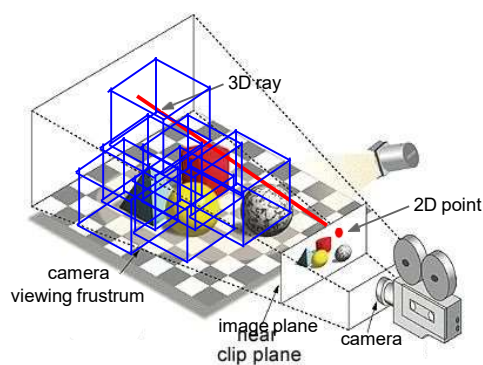
Much harder problem than 2D object detection!



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3D Object Detection: Simple Camera Model



A point on the image plane corresponds to a **ray** in the 3D space

A 2D bounding box on an image is a **frustum** in the 3D space

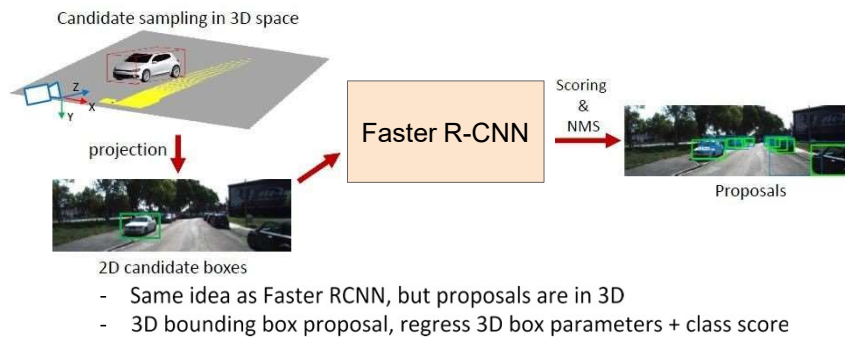
Localize an object in 3D:
The object can be anywhere in the **camera viewing frustum**!



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3D Object Detection: Monocular Camera



Chen, Xiaoqi, Kaustav Kundu, Ziyu Zhang, Huimin Ma, Sanja Fidler, and Raquel Urtasun. "Monocular 3d object detection for autonomous driving." CVPR 2016.

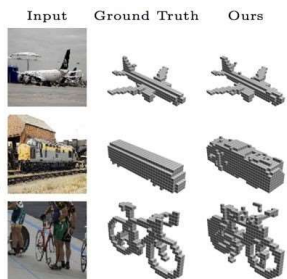


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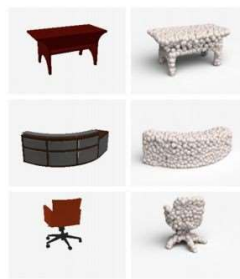
3D Shape Prediction

Voxel:
D x D x D binary



Choy et al, "3D-R2N2: A Unified Approach for Single and Multi-view 3D Object Reconstruction", ECCV 2016

Pointcloud:
V x 3 float



Fan et al, "A Point Set Generation Network for 3D Object Reconstruction from a Single Image", CVPR 2017

Mesh:
V x 3 float, F x 3 int



Wang et al, "Pixel2Mesh: Generating 3D Mesh Models from Single RGB Images", ECCV 2018



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Open Source Frameworks

- Lots of good implementations on GitHub!
- TensorFlow Detection API:
 - https://github.com/tensorflow/models/tree/master/research/object_detection
 Faster RCNN, SSD, RFCN, Mask R-CNN
- Caffe2 Detectron:
 - <https://github.com/facebookresearch/Detectron>
 Mask R-CNN, RetinaNet, Faster R-CNN, RPN, Fast R-CNN, R-FCN
- Finetune on your own dataset with pre-trained models



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References

1. CS231n: Convolutional Neural Networks for Visual Recognition
<http://cs231n.stanford.edu/>
2. Object Detection Creation from Scratch. Samsung R&D Institute Ukraine. Vitaliy Bulygin
<https://aiukraine.com/wp-content/uploads/2018/08/Vitalij-Buly-gin-.pptx>
3. CornerNet: Detecting Objects as Paired Keypoints
<https://pvl.cs.princeton.edu/assets/CornerNet.pptx>



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